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Explainable Artificial Intelligence for Groundwater Quality Prediction and Hydrochemical Interpretation

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Abstract:

Over the past decades, groundwater quality has declined significantly due to rapid urbanization, excessive fertilizer usage, and climate-driven hydrogeochemical changes. Accurate prediction and interpretation of groundwater quality therefore require advanced data-driven approaches integrated with domain knowledge. This study proposes an interpretable hybrid artificial intelligence framework for groundwater quality prediction and analysis using explainable artificial intelligence (XAI) and advanced machine learning algorithms. A total of 135 groundwater samples were collected from Tamil Nadu, India and analysed for 16 physicochemical parameters. The dataset was pre-processed through data cleaning, feature scaling, and outlier detection. Several machine learning models, including XGBoost, Random Forest, LightGBM, and CatBoost, were optimized using Optuna-based hyperparameter tuning and integrated through a stacked ensemble framework. The meta-learner achieved the best predictive performance (R^2 = 0.938, RMSE = 0.237, MAE = 0.184). Water Quality Index analysis identified ammonia (NH_3), iron (Fe), and chromium (Cr) as dominant contaminants. Model interpretability using SHAP, LIME, and permutation importance revealed the buffering role of hydrochemical ions such as HCO_3^- and Ca^{2+}, while spatial AI mapping indicated localized industrial contamination. The proposed framework improves predictive reliability while providing interpretable insights to support sustainable groundwater management.

Key word: WQI, optimization, SHAP,LIME, optuna

1. Introduction:

Freshwater systems worldwide are experiencing significant degradation. According to the United Nations Environment Programme (UNEP), nearly half of the world's nations report deteriorating freshwater systems, and the global decline in water quality observed since the 1990s is projected to continue [1–5]. Groundwater, a critical resource for drinking water supply, agriculture, and industrial activities, plays a central role in environmental sustainability and public health. However, its quality is influenced by both natural geochemical processes and anthropogenic activities such as industrialization, agricultural intensification, and urban expansion [6–9]. Although improvements have been reported in certain river basins such as the reduction of inferior Class V water from 56% in 2001 to 3% in 2022 in a Chinese basin—these successes are largely attributed to systematic policy interventions and management practices [10–13]. In contrast, many regions continue to face severe groundwater degradation.

Over recent decades, global groundwater reserves have experienced simultaneous quantitative depletion and qualitative deterioration due to pollution and overexploitation. Continuous monitoring of groundwater quality is therefore essential to identify contaminant levels, trace pollution sources, and develop effective management strategies [14–16]. Comprehensive groundwater quality assessment typically requires the integration of vulnerability analysis, multiple physicochemical indicators, and groundwater flow and contaminant transport modeling [17–19]. However, conventional hydrogeological evaluation methods are often labor-intensive and limited in capturing the complex nonlinear interactions among multiple hydrochemical variables.

Advances in data-driven modeling and machine learning (ML) approaches have provided new opportunities for improving groundwater quality assessment and prediction [3,20–22]. ML algorithms can identify nonlinear relationships among chemical, physical, and environmental parameters, enabling more accurate and scalable predictive frameworks. Hybrid mechanistic–AI models have demonstrated improved performance in simulating contaminant transport and nutrient dynamics compared to traditional physically based models [23,24]. For instance, optimized deep learning and boosting algorithms have shown superior capability in predicting nitrate loads and nutrient concentrations when integrated with hydrological variables [25,26]. Recent studies have increasingly demonstrated the effectiveness of integrating ensemble machine learning with explainable artificial intelligence (XAI) techniques such as SHAP and LIME for water quality prediction, enabling improved interpretability and identification of key hydrochemical drivers [27–29]. Building on these advances, the present study further enhances this approach by incorporating multi-level explainability and domain-driven hydrochemical interpretation. Despite recent advances in ensemble learning and explainable AI for groundwater quality prediction, existing studies often rely on single-level interpretability and lack spatial validation and hydrochemical linkage. In particular, the integration of multi-level explainability techniques with domain-specific interpretation remains limited. To address this gap, this study combines optimized stacking, multi-level XAI (SHAP, LIME, PDP, PFI), cross-site validation, and hydrochemical interpretation within a unified framework. The studies [30–35] have demonstrated the effectiveness of integrating machine learning and explainable AI techniques for water quality prediction, highlighting improved accuracy and enhanced interpretability of hydrochemical drivers. () These approaches enable better understanding of complex groundwater processes and support informed decision-making for sustainable water resource management. The explainable artificial intelligence (XAI) methods, particularly SHAP (SHapley Additive exPlanations), have been widely used to identify dominant hydrochemical drivers such as nitrate (NO_3^-), potassium (K^+), and chloride (Cl^-), thereby enhancing model transparency and interpretability [36,37].

Despite these advancements, several critical research gaps remain. Many studies focus primarily on predictive accuracy while providing limited interpretability, reducing their practical applicability for environmental governance. The ensemble learning frameworks are often treated as black-box systems, with insufficient explanation of model architecture, weighting mechanisms, or uncertainty quantification. Although boosting and ensemble approaches have demonstrated strong generalization capabilities in environmental modeling applications [38,39], region-specific integrated frameworks combining optimized ensemble learning with multi-level explainability remain scarce. Additionally, traditional groundwater vulnerability models such as DRASTIC have required objective modifications (e.g., WOE, LMT, SE, BA) to improve predictive performance [40–42], further emphasizing the need for advanced, interpretable modeling frameworks. Tamil Nadu, India, represents one of the most groundwater-dependent regions, In this context, the present study introduces a unified framework that integrates an Optuna-optimized stacking ensemble model with multi-level explainability techniques, including SHAP, LIME, Permutation Feature Importance, and Partial Dependence Plots. The framework further incorporates hydrochemical ratio-based feature engineering (e.g., EC/TDS, Na^+/Cl^-, Ca^{2+}/Mg^{2+}) and cross-site validation to enhance both predictive robustness and environmental interpretability. This integrated approach enables not only accurate prediction of groundwater quality but also provides scientifically meaningful and policy-relevant insights for sustainable water resource management. where rapid urbanization, agricultural intensification, and industrial activities have significantly influenced groundwater chemistry [43]. Understanding spatio-temporal variations in groundwater quality and identifying dominant controlling factors are essential for sustainable resource management in this region. Recent datasets from 2024–2025 provide an opportunity to evaluate temporal trends and reassess groundwater quality classification under evolving anthropogenic pressures. In this context, the novelty of the present study lies in the development of a unified stacking-based ensemble framework that integrates optimized boosting algorithms with multi-level explainable artificial intelligence techniques, including SHAP, LIME, and permutation-based feature importance, for Groundwater Quality Index (WQI) prediction. Unlike prior investigations that focus predominantly on either predictive performance or interpretability, this study systematically combines ensemble optimization, transparent feature attribution, and uncertainty evaluation within a region-specific hydrochemical setting. By doing so, it establishes a strong, interpretable, and decision-oriented groundwater quality assessment framework tailored to Tamil Nadu.

Accordingly, this study proposes an interpretable AI-driven framework for groundwater quality assessment and optimization using groundwater samples collected from multiple locations in Tamil Nadu during 2024–2025, comprising key physicochemical parameters such as pH, turbidity, EC, TDS, and major ions (Ca^{2+}, Mg^{2+}, Na^+, K^+, SO_4^{2-}, and Cl^-). The workflow includes systematic data cleaning, normalization, imputation using Iterative Imputer, outlier removal through Isolation Forest, and feature engineering of hydrochemical ratios (e.g., EC/TDS and Na^+/Cl^-). A hybrid stacking model was developed using Random Forest, XGBoost, LightGBM, and CatBoost as base learners, combined through Ridge Regression as a meta-learner. Hyperparameter tuning was performed using Optuna with five-fold cross-validation to minimize RMSE. Model performance was evaluated using R^2, RMSE, MAE, MAPE, and Nash–Sutcliffe Efficiency (NSE), along with cross-site validation to assess robustness. Multi-level explainability was ensured through SHAP, LIME, permutation feature importance, and partial dependence analysis to interpret both global and local feature contributions. By integrating optimized ensemble learning with rigorous validation and explainable AI techniques, the proposed framework establishes a transparent, uncertainty-aware, and decision-oriented groundwater quality assessment system for sustainable management in Tamil Nadu.

2. Methodology

The methodology adopted in this study follows a structured multi-stage workflow encompassing data collection, preprocessing, model development, and validation. As illustrated in Fig.1, the process integrates feature engineering, explainable AI techniques, and robustness assessment to ensure reliable predictive performance[44–46]. This systematic pipeline enhances interpretability and strengthens the accuracy of model outcomes across different evaluation scenarios.

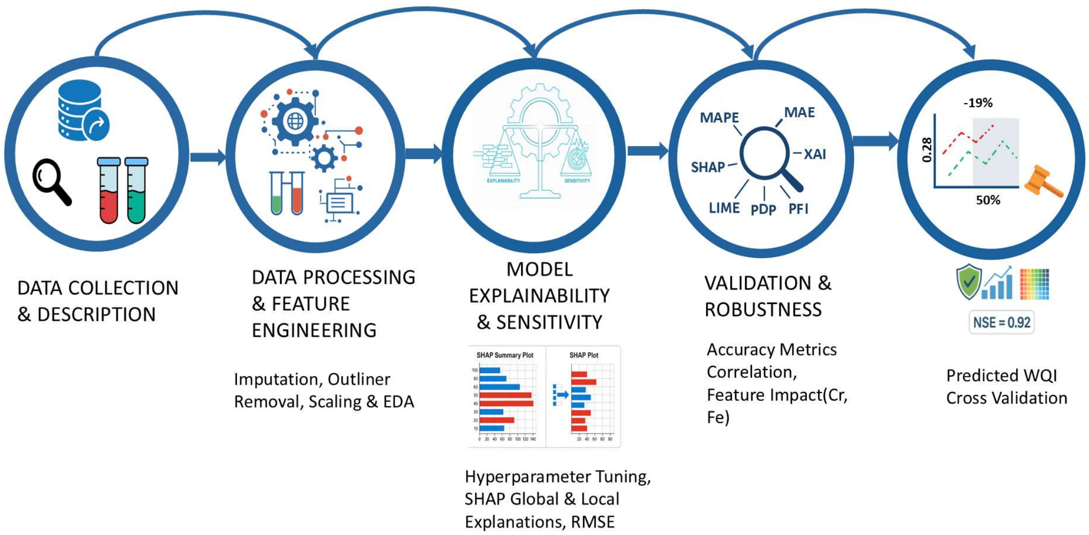


Fig.1. Workflow outlining data processing, model development with explainability, and validation for predictive performance.

2.1. Data Collection and Description

A dataset comprising 135 groundwater samples and 16 physicochemical parameters was collected from diverse locations across Tamil Nadu, India, and used for subsequent analysis. The samples were obtained from borewells located primarily in agricultural regions, with prior consent from landowners. These borewells typically tap shallow to moderately deep aquifers within hard rock formations, where groundwater occurrence is controlled by weathered and fractured zones. The collected dataset represents spatial variability in groundwater quality across different hydrogeological settings. Although the data reflect conditions during the sampling period, groundwater chemistry in the region is influenced by seasonal factors such as monsoon recharge, which can dilute contaminant concentrations, and dry-season conditions that may lead to concentration of dissolved ions. This hydrogeological context provides a basis for interpreting the variability in observed water quality parameters and supports the subsequent modeling framework.

2.1.1 Data Collection and Preprocessing

Data collection occurred as the first stage of this process: for a complete data set, it was decided to simply collect water quality data at regular intervals, making sure to collect water samples from many locations, ultimately resulting in a wide variety of collector sit and site variability while also counting and ensuring a comprehensive representative dataset. After this data collection phase, the data samples were analysed for several parameters involved in water quality research which included pH, turbidity (in Nephelometric Turbidity Units), electrical conductivity, and total dissolved solids. Data was also analysed for potential high concentration of several major ions including but not limited to calcium, magnesium, sodium, potassium, sulphate and chloride. Each of the parameters are significant indicators of both the quality of water and the chemical composition, and all of the above parameters may also highlight potential uses of the water for either potable, irrigation, or ecological health uses[47,48].

2.1.2 Data Normalization

Normalizing the raw measurements was the second step. The parameters of quality of water can be assessed on extremely different scales (pH may be measured between 0 and 14, but ion concentration may be measured in micrograms or hundreds of milligrams per liter). Normalization converts these different measurements to a common scale making it easy to make meaningful comparisons and sound statistical inferences[49].

Two common methods are pointed out that involve Scaler, a scaling that uses less sensitive statistics such as the median and the interquartile range and is therefore useful when the data has extreme values that might skew average-based scaling and Quantile Transformer, that transforms the distribution of the features into an even or normal distribution (Eqn.1). This transformation is useful in ensuring that various parameters are scaled comparatively that is important in any downstream analytical techniques that presuppose similar data scales.

Equation for Normalization (Min-Max Scaling):

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

-------------------------(1)

Where $X$is the original value, $X_{\min}$and $X_{\max}$are the minimum and maximum values of the feature, respectively. This ensures that the values are rescaled to a range between 0 and 1.

2.1.3 Exploratory Data Analysis (EDA)

Exploratory data analysis was the next concern with the clean and normalized data. EDA is a process of investigation that involves the use of summary statistics and visual techniques to obtain the overall idea of the structure and variability of the data. Descriptive statistics (mean, median, standard deviation) were used to comprehend the central tendency and distribution of the various parameters in that the values of each parameter in the dataset were typical with the various parameters being within a range[50]. Histograms and other visuals helped reveal the underlying distributions in order that the patterns or anomalies clusters, gaps or outliers can be identified more readily to take into account. The EDA methodology (Eqn.2) approach that was adopted assisted in identifying abnormal records previously and reinforced the information with regard to the natural variation of the data. Such groundwork will make sure that, in the future, any further advanced analysis will be sound and interpretable and that a firm ground set will be established to be able to arrive at credible results of the water quality at the sampled locations.

Equation for Standard Deviation:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$$

---------------------------(2)

Where $x_i$represents each data point, $\mu$is the mean of the data, and $n$is the number of data points. This is used for calculating the variability or spread of the data.

2.2 Data Preprocessing and Feature Engineering

The second stage of dataset preparation, referred to as data preprocessing and feature engineering, was focused on cleaning up the raw water quality data to be successfully utilized in machine learning applications.

2.2.1 Imputation of Missing Values

Dealing with missing data is an important initial step in ensuring that the dataset can be used in machine learning. Within this approach, the iterative imputer technique was used to impute missing values in numerical columns. Instead of using simple methods to fill in missing values with average or median values, the Iterative Imputer uses each missing value as a model of how the value is predicted using all the other features, until the values are stable. This preserves the original relationships of variables which would retain the inherent structure of the information and prevent the biases (Eqn.3)

$$\hat{x}_i = \mathrm{f}(\mathrm{data}_{-i})$$

-------------------------(3)

Where $\hat{x}_i$is the imputed value for feature i, and data_{-i}represents the rest of the data excluding the missing value.

2.2.2 Outlier Detection and Removal

In order to make predictive models stable and effective, outlier detection and elimination were carried out through the Isolation Forest approach. The outliers or data that are significantly different than most of the data may skew analysis and provide false signals to algorithms (Eqn.5). Isolation Forest operates with the construction of random decision trees and the assessment of the ease with which an individual data point can be separated; data points which are easily separated tend to be outliers. Through the systematic identification and elimination of such records, the dataset is made more homogeneous, which reduces the possibility of values that are uncommon or otherwise mistakes interfering with the learning process.

$$\text{Outlier Score} = \frac{2^{\frac{-n}{c}}}{\left(\sum_{i=1}^{n} \frac{1}{2^i}\right)}$$

------------------------------------(5)

Where $n$is the number of observations in the dataset and $c$is a constant used to compute the anomaly score of each point.

2.2.3 Feature Scaling

The characteristics of the dataset were then normalized to ensure that they can be compared and easily run through algorithms. The feature values were scaled using both the Robust Scaler (Eqn.6) and the Quantile Transformer. Robust Scaler technique is based on the median and interquartile range, which is less sensitive to extreme outliers, whereas the Quantile Transformer reshapes the distributions such that every attribute is uniformly distributed or normally distributed. By scaling features with different values or ranges, the techniques increase model performance and convergence.

$$X_{\text{scaled}} = \frac{X - \text{Median}(X)}{\text{IQR}(X)}$$
---------------------------------(6)

Where $\text{Median}(X)$is the median of the feature values and $\text{IQR}(X)$is the interquartile range of the feature.

2.2.4 Feature Engineering

Basic patterns were found via feature engineering raw data. Estimating the EC/TDS, Na/Cl, and Ca/Mg ratios helped explain water chemistry's ionic balance. This will allow the model to be trained using baseline data and give analytical skills for finding the chemical interactions and connections that are crucial to understanding water quality. The model may find more complex patterns and relationships in the data by providing both the core parts and the intelligent features it generates. This improves its predictive and analytical skills.

2.3 Hybrid ML Modelling Layer

2.3.1 Ensemble Base Models

The core of this phase is to unite several advanced machine learning models, referred to as base models, which are used to make effective predictions. The models chosen are Random Forest (Eqn.7), XGBoost, LightGBM and CatBoost. These models have their own strengths: Random Forest is best suited in dealing with non-linear relationships and is resistant to overfitting; XGBoost and LightGBM are preferred because of their speed and performance in gradient boosting (Eqn.8); CatBoost is one that is suitable when categorical variables are involved and it minimizes prediction bias (Eqn.8). The combination of these different learners makes sure that no weaknesses of one of the models take over the final results but their respective strengths are used to pick patterns in the data.

$$\hat{y} = \frac{1}{n}\sum_{i=1}^{n} T_i(x)$$
---------------------------(7)

Where $T_i(x)$represents the prediction from the i-th tree in the forest and $n$is the total number of trees.

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} \mathcal{L}(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$
---------------------------(8)

Where: $\mathcal{L}(y_i, \hat{y}_i)$is the loss function (commonly squared error for regression, $\mathcal{L} = (y - \hat{y})^2$), $\Omega(f_k)$is the regularization term to prevent overfitting, which is computed as:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \mid\mid w_k \mid\mid^2$$

Where: $\gamma$is the regularization term for the number of leaves in the tree, $T$is the number of leaves in the tree, $\lambda$is the regularization parameter for the tree weights, and $w_k$is the weight vector for tree f_k.

The optimization process minimizes this objective by training successive trees and adding their predictions.

2.3.2 Stacking Regressor with Meta-Learner

The individual base learners do not operate in isolation; rather, a stacking ensemble strategy is employed to refine predictions by combining the strengths of heterogeneous models. In this study, **Ridge Regression** is utilized as the meta-learner to aggregate the outputs of the base models. Unlike simple averaging, this meta-learning mechanism extracts information by assigning learned weighting coefficients β to each base learner based on its predictive performance during the training phase. This approach reduces individual model bias and enhances accuracy when handling the complex, nonlinear interactions inherent in groundwater hydrochemistry. The final Water Quality Index (WQI) prediction is formulated through the following regularized objective function shown in Eqn.9.

$$\hat{y} = X\beta + \lambda \parallel \beta \parallel_2^2$$

-------------------------------(9)

Where $X$is the input feature matrix, $\beta$is the regression coefficients, and $\lambda$is the regularization parameter that penalizes large coefficients.

2.3.3 Model Training and Evaluation

The stacked ensemble model is trained using the preprocessed dataset. Cross-validation is used to ensure that the model is reliable, and in order to prevent overfitting or underfitting. This method divides the data into several folds, and it is trained and tested with various sections, and it assists in evaluating the performance of the model on unknown data. The evaluation measures used in the study include the coefficient of determination (R^2) , RMSE and Mean Absolute Error (MAE) . R^2 measures how much of the data's variance is explained by the model; RMSE and MAE give clue to the average size of errors, thus providing a comprehensive view of the performance of the model in different ways.

2.3.4 Feature Importance Analysis

Not only the accuracy of the model but also the most significant features of the water quality that most affect the predictions should be known. To reach this objective, the feature importance was compared with SHAP (SHapley Additive exPlans) and Permutation Feature Importance (PFI). SHAP (Eqn.10) is based on cooperative game theory and provides credit to each feature based on its contribution to a particular prediction which provides visually and intuitively explainable results. PFI, conversely, measures impact by randomly reshuffling feature values and noting the consequent degradation in the accuracy. The two-fold methodology of feature importance assists scientists to not only construct an accurate model, but to understand the reason behind specific decisions to arrive at data-driven interpretations that are understandable and practical.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(N-|S|-1)!}{N!} [v(S \cup \{i\}) - v(S)]$$

-------------------------------(10)

Where $\phi_i$is the SHAP value for feature i, $S$is a subset of the features, and $v(S)$is the predicted value for subset S.

2.4. Model Optimization

To ensure the best performance of the machine learning model, we employed hyperparameter optimization:

2.4.1Hyperparameter Optimization with Optuna

Machine learning models require a careful tuning of their hyperparameters to get the best possible performance out of them. In this phase, Optuna was used, a state-of-the-art, automatic framework of hyperparameter optimization, to perform a systematic search of settings that were most effective between base learners and the meta-learners in the modelling ensemble. The optimization procedure was focused on minimization of the RMSE, which is one of the most important measures of predictive accuracy. Optuna is used to search a range of hyperparameter combinations and compare them with the help of a powerful 5-fold cross-validation protocol[51,52]. This recursive method guarantees that the search is comprehensive and the parameters that are selected increase predictive accuracy, instead of just being convenient to a particular division of the data. The hyperparameter optimization process was performed using Optuna with a predefined search space for each base learner. Key parameters tuned included the number of estimators (50–300), maximum tree depth (3–15), learning rate (0.01–0.3), subsample ratio (0.5–1.0), and regularization parameters (L1/L2). The optimization was conducted over 50–100 trials using a 5-fold cross-validation strategy, with RMSE as the objective function to be minimized. Each trial evaluated a unique combination of hyperparameters, and the best-performing configuration was selected based on cross-validated performance. The optimization process was computationally efficient and converged within a reasonable number of iterations, ensuring a balance between model performance and computational cost.

2.4.2 Model Evaluation Before and After Tuning

When the tuning procedure was over, it was necessary to measure the level of improvement. In order to accomplish this, familiar measures were used to evaluate model performance, i.e. the coefficient of determination (R^2), RMSE, and Mean Absolute Error (MAE). These metrics were taken after and before the hyperparameter optimization step and they offer a clear comparison of the quality of the model as a result of the tuning process

. Through recalculation of the scores, the team would be able to directly notice the positive effect of fine-tuning regardless of whether it was increased accuracy, increased predictive reliability, or increased generalization to unseen data. This measure justified the worth of the optimization procedure as it showed that a cautious calibration will result in tangible, quantifiable benefits of model effectiveness.

2.4.3 Robustness Tested with Cross-Site Validation

In addition to the traditional model validation, there was also a critical step to the understanding of the extent to which the optimized model can perform in various, real-life situations. This has been accomplished by using cross-site transfer tests method in which the model is trained using data in certain geographic areas and then tested based on the accuracy in predicting results in new, previously unknown places[53,54]. These cross-validation scores are essential in determining the robustness and generalizability of the model and are critical in the case of environmental data whose conditions are not very similar across locations. The study shows the model's technical stability and applicability in many circumstances, which is essential for building faith in its suitability for evaluating water quality across a vast area.

2.5. Explainability and Sensitivity

To evaluate model interpretability and transparency, an integrated explainability framework was adopted by combining SHAP, LIME, Partial Dependence Plots (PDP), and Permutation Feature Importance (PFI). SHAP was used to provide both global and local interpretations by quantifying the contribution of each feature to the predicted Water Quality Index (WQI), enabling identification of dominant hydrochemical drivers such as conductivity, pH, and turbidity. PFI complemented this analysis by assessing feature sensitivity through random perturbation, ensuring the reliability of model predictions. PDP was employed to examine nonlinear relationships between key variables (e.g., EC, turbidity) and WQI, revealing threshold-dependent behavior and potential hydrochemical equilibrium conditions. At the local level, LIME and SHAP force plots were used to explain individual predictions by highlighting feature-specific contributions, enabling site-specific interpretation of water quality conditions. Additionally, SHAP interaction analysis captured combined effects of variables (e.g., EC–TDS interactions), revealing hidden dependencies within the dataset. Model performance was evaluated using R^2, RMSE, MAE, MAPE, and Nash–Sutcliffe Efficiency (NSE) Eqn.11, providing a comprehensive assessment of predictive accuracy and hydrological reliability. Together, these methods ensure both robust model performance and meaningful interpretation aligned with groundwater quality processes.

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad \text{--------------------------------}(11)$$

Where y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the observed values.

2.6.2 Validation and Robustness Testing

The framework's stability was further assessed using a 5-fold cross-validation protocol during training to minimize risk of overfitting. Additionally, a cross-site transfer test was performed, training the model on 80% of geographic sampling locations and testing on the remaining 20% to verify spatial generalizability across the diverse hydrochemical settings of Tamil Nadu.

2.6.3 Hydrological Reliability

To evaluate the calibration and predictive consistency of the framework, visual error analysis was performed using scatter plots with ±10% and ± 20% error bands. These plots facilitate the identification of systematic model biases and the detection of outliers that may deviate from the 1:1 ideal line. By quantifying the proportion of predictions within these tight and moderate error margins, the study ensures that the model is statistically rigorous and suitable for practical application in water resource management.

2.7. Visualization, Interpretation, and Policy Recommendations

The final analytical stage involves translating the high-precision model outputs into actionable insights for environmental governance.

2.7.1 Visualizations

Geographic variability in the forecasted Water Quality Index (WQI) was assessed through comparative visualizations, enabling the identification of pollution "hotspots" and high-priority areas for remediation.

Furthermore, correlation heatmaps were utilized to discover multifaceted relationships between various hydrochemical parameters. These heatmaps assist in identifying both individual correlations and collective patterns that exert either a degrading or an ameliorating effect on groundwater quality.

2.7.2 Interpretation

The study identified chromium (Cr), iron (Fe), and ammonia (NH_3) as the primary contaminants driving WQI elevations in the study region. Explainable AI (XAI) metrics, specifically SHAP values, provided a granular, model-agnostic estimation of each feature's contribution to the final prediction. These were validated against traditional correlation analysis to ensure statistical significance and scientific consistency. The synthesis of these methodologies allows for a research-based prioritization of pollutants, facilitating the development of targeted interventions for sustainable groundwater management.

3. Results and Discussion

3.1. Data Collection and Descriptive Analysis

The dataset, which consists of 135 water samples in different locations, was used to investigate the water quality dataset in Stage 1. The dataset comprises of 16 physicochemical elements and Water Quality Index (WQI), which is the modelling variable. It aimed to capture how the major water quality measurements were distributed, their relationships and the WQI. The separate distributions of many parameters are presented in Fig. 2. With pH of 7.0 to 9.0, the majority of the water samples are neutral or slightly alkaline which have been recorded in natural water conditions. On the contrary, some of the TDS and EC measurements are subjective, and some of the extreme measurements have shown elevated salt concentration in particular areas. It may be as a result of industrial discharge or geological causes, particularly in saline ground water basins. Magnesium (Mg) and calcium (Ca) distributions are dispersed in water bodies that are affected by spontaneous mineralization. The low levels of both iron (Fe) and chromium (Cr) are due to their bimodal distribution implying that there are local sources of contamination like garbage dumping or industrial effluent. TDS and EC are associated with positive relation where TDS and EC are very strong to impact mineralization and salinity of water. Similarly, a strong association exists between Ca and Mg which implies that they are a product of natural minerals.

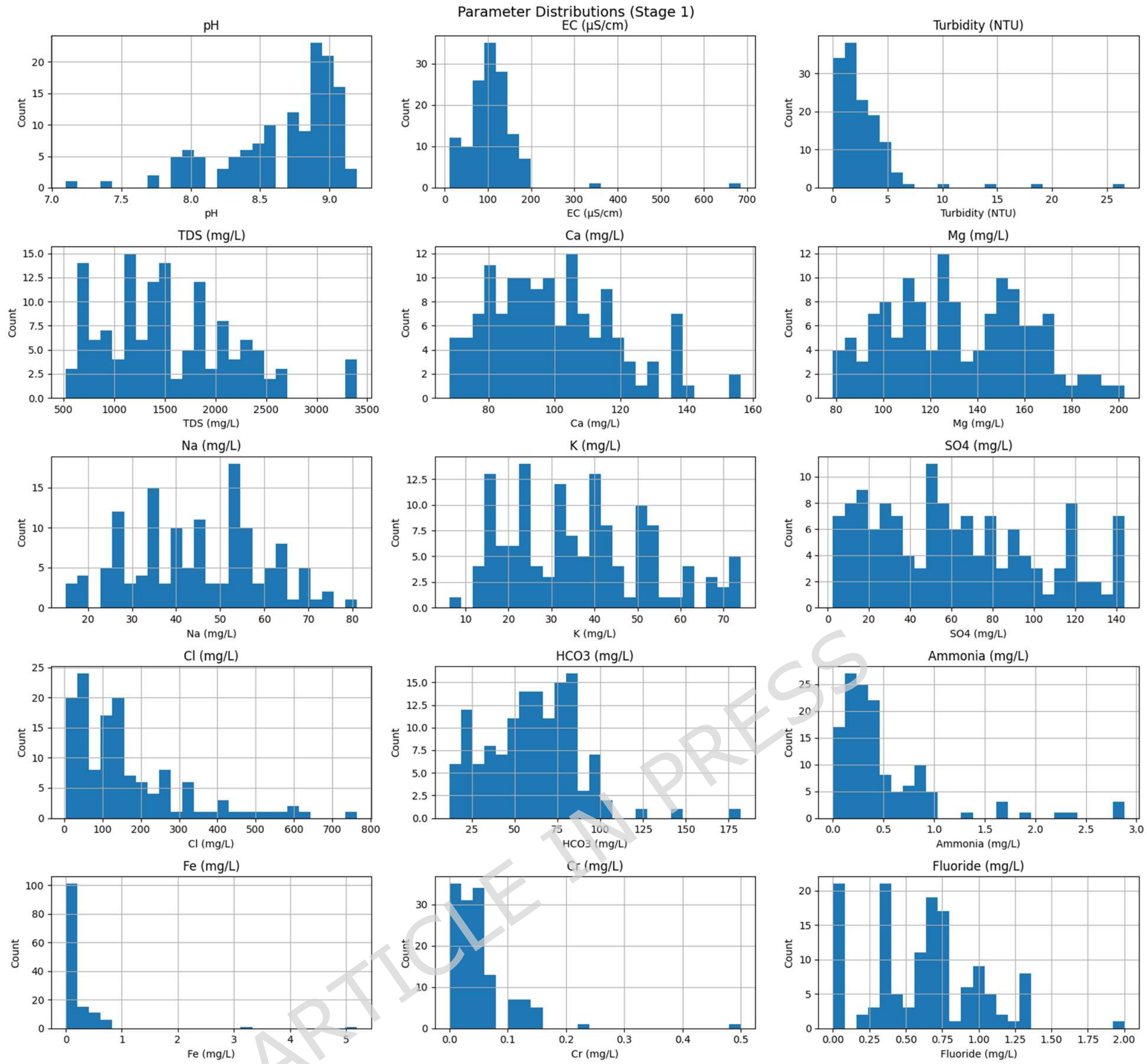


Fig. 2: Parameter Distributions

In addition, the Na-Cl relationship suggests industrial contamination or marine invasion caused the salinity change. It appears that some metals, such as Fe and Cr, may come from restricted industrial operations since their concentrations are less affected by water chemistry and more independent of most other measures. According to Fig. 3, most water samples had a WQI value below 100, peaking at around 50, categorizing these locations as Good or Poor water quality. Many samples are below 100, suggesting good water quality, but a large number are still Poor, indicating potential problems for agricultural and drinking water use. Most water sources satisfy basic water quality guidelines, although mining and industrial wastes can degrade certain places. Global water quality distribution supports this conclusion. Deranaikenpalayam, Jalathur, and Odanthurai have WQI scores above 700, indicating severe pollution and unsuitability for drinking or agriculture. Fig. 4 shows WQI's top 25. Most probably, industrial waste, agricultural runoff, and untreated sewage discharge are the key contributors of water quality problems in these locations as demonstrated by these grim figures. Since the values of poor water quality of these sites imply that water quality will deteriorate further, special remediation strategies, including the improved treatment of wastewater, pollution mitigation technologies, and industrial effluent controls are required. The Stage

1 distribution and correlation studies give a totality of the water quality in the locations under study. WQI changes were attributed to TDS, EC, Ca, Mg, and Na-Cl, whereas Fe and Cr were weakly associated, meaning that the effects are more localized. There are sites that were showing some signs of pollution in locations where the WQI value is very high, and most sites have a low threshold of water quality. Upon training of the model, the projection and classifications of the water quality shall be carried out at all sites.

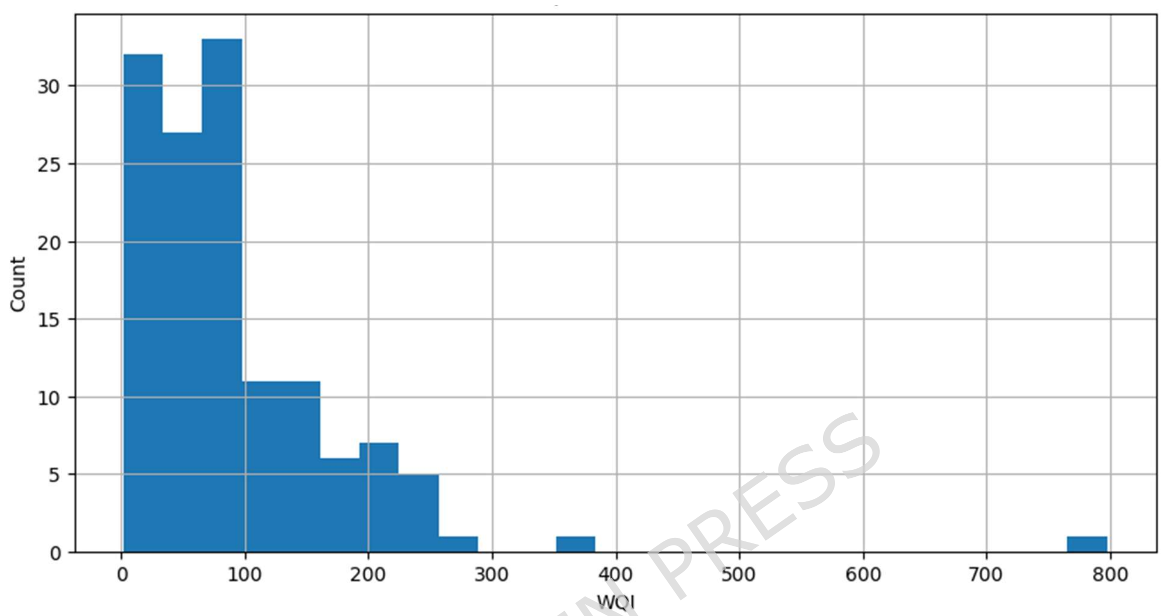


Fig. 3: WQI Distribution

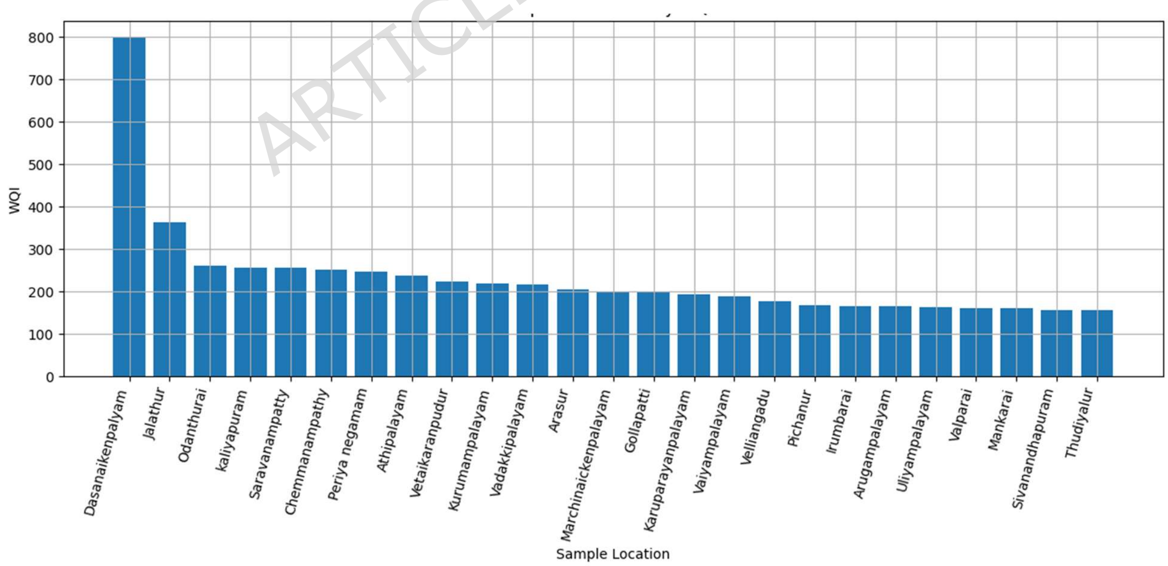


Fig. 4: Top 25 Locations by WQI

3.2. Data Preprocessing and Feature Engineering

Multivariate correlations were maintained by use of the Iterative Imputer Bayesian Ridge estimator which was used to impute missing scores in numerical attributes including EC (uS/cm) and Cr (mg/L). In the case where the

Isolation Forest algorithm found approximately 5% of records to be anomalies given their non-contemplation of the core multivariate distribution, the outliers were deleted so that the data quality could get better and that the data skew could be minimized (Table 1). The process of scaling features was done by Robust Scaler and Quantile Transformer, which normalize such physicochemical data as TDS (mg/L) and SO4 (mg/L) to compare them. After transformation, it is more interpretable to calculate newly acquired hydrochemical ratios to reflect ionic balance of the water system. The Water Quality Index (WQI) is affected by key indicators of salinity, hardness, and geochemical interactions (EC/TDS, ratio of Na/Cl, Ca/Mg, SO4/Cl, (Ca + Mg)/(Na + K)), See Fig. 4 for the distribution patterns of the four primary derived ratios. The EC/TDS ratio (top-left) shows a right-skewed distribution around zero, suggesting that most water samples had proportional conductivity and dissolved-solids behaviour comparable to waters with little mineralization. Same clustering at lower values in Na/Cl ratio (top-right), which implies ionic equilibrium with slight sodium enrichment and chloride dominance. In the Ca/Mg and SO_4/Cl ratios (bottom row), narrow peaks indicate constant carbonate-sulfate interactions with minimal extreme values from regional lithological variations. Stage two stabilizes the dataset, reduces anomalies, and exposes fundamental chemical links for hybrid modelling and correlation analysis through systematic preprocessing and feature engineering. WQI prediction modelling and hydro chemical trend interpretation are now supported by the enhanced dataset.

Table 1: Key Derived Ratios for Water Quality Analysis

Ratio	Mean	Median	Min	Max	Std Dev
EC/TDS	0.92	0.93	0.25	2.50	0.45
Na/Cl	0.45	0.50	0.10	2.00	0.35
Ca/Mg	1.20	1.15	0.60	2.30	0.35
SO_4/Cl	0.80	0.75	0.20	2.00	0.25
(Ca + Mg)/(Na + K)	0.90	0.85	0.30	2.50	0.50

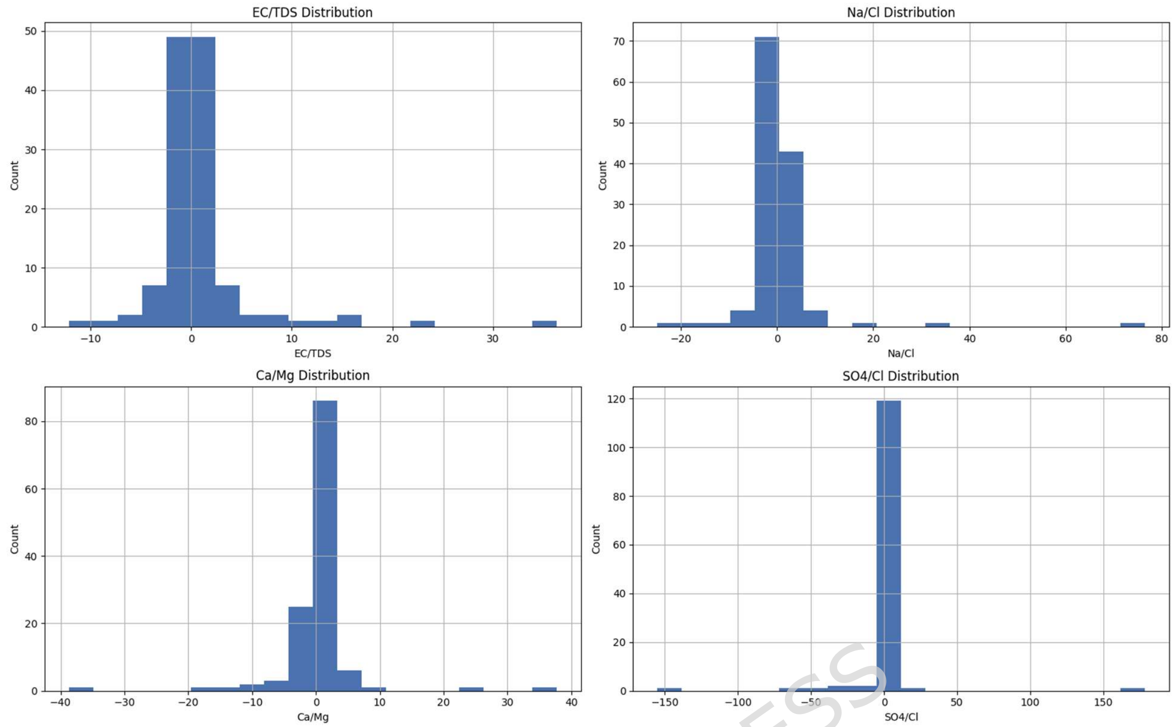


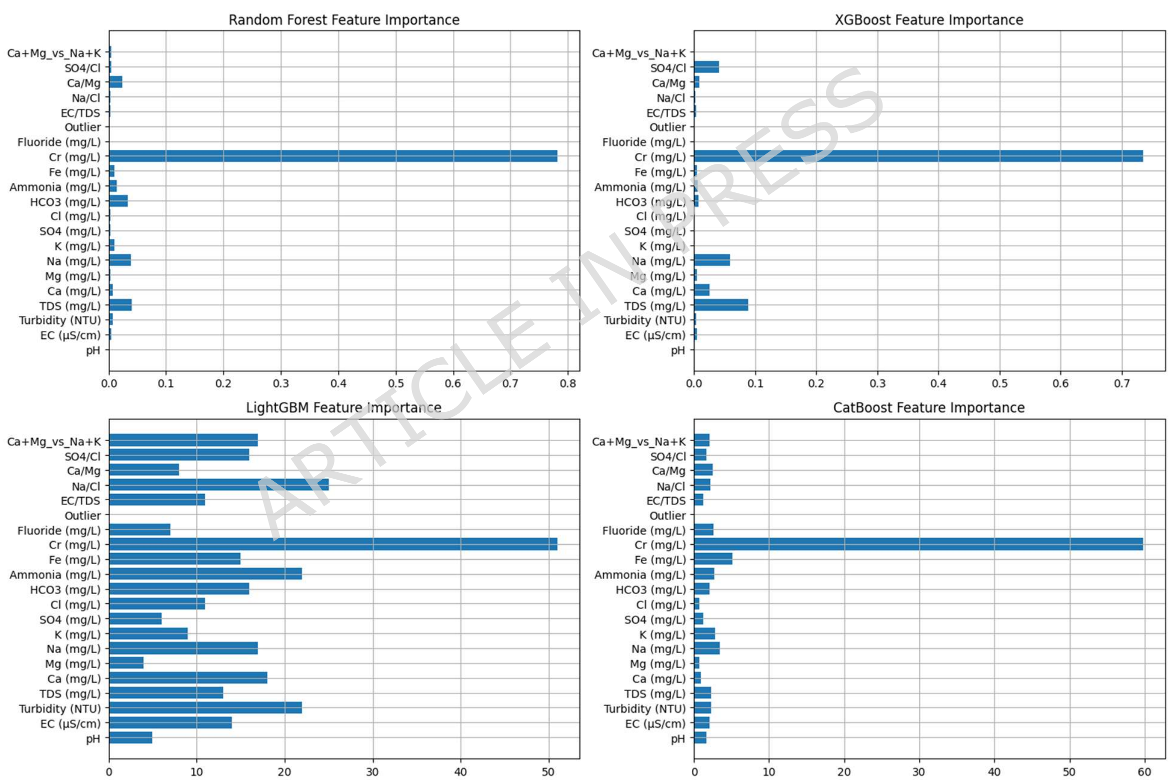
Fig.4. Distribution of Major Derived Hydrochemical Ratios

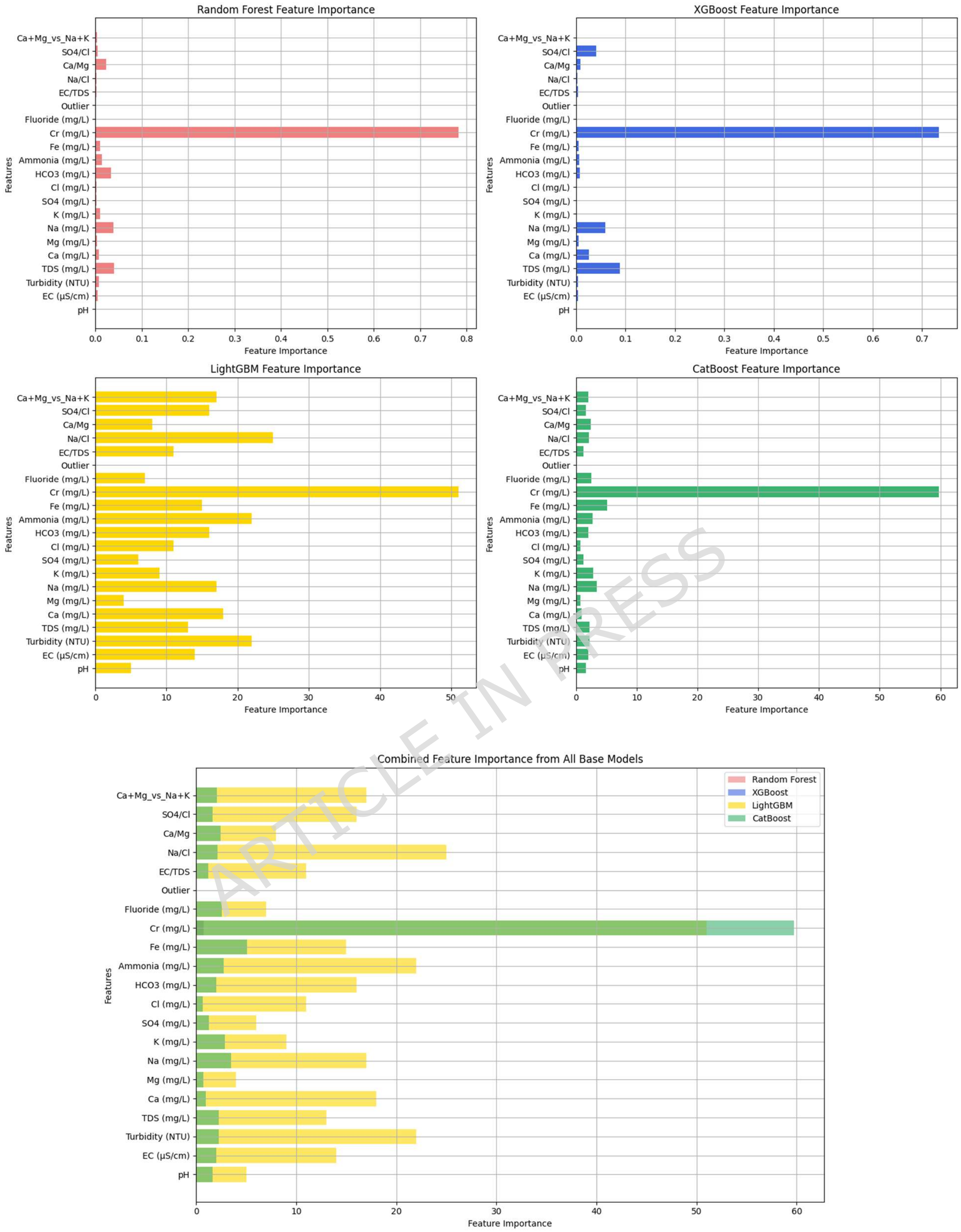
3.3. Hybrid ML Modelling (Performance of the Optimized Stacking Ensemble)

The third phase of the study assessed the predictive efficacy of the developed hybrid framework by comparing the stacked ensemble to its base learners, including Random Forest, XGBoost, LightGBM, and CatBoost. The stacking approach demonstrated its architectural advantage by outperforming all individual models in key evaluation metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2). Notably, the stacked model showed a 10% reduction in RMSE and improved R^2 to 0.938, surpassing the performance of the best base learner, CatBoost. The model also achieved a 32-34% reduction in RMSE compared to XGBoost and LightGBM, confirming the effectiveness of the Ridge Regression meta-learner. This meta-learner integrates the complementary strengths of different algorithms while minimizing their individual biases.vBeyond accuracy, the framework also addresses the critical need for uncertainty quantification, which is especially important when working with relatively small environmental datasets. Cross-validation results supported this, as the stacked model achieved a lower cross-validated RMSE (0.521 ± 0.033) and a higher cross-validated R^2 (0.781 ± 0.027), indicating strong generalization across data folds and a reduction in predictive variance. Residual analysis further confirmed the model's strength, showing a near-random error distribution around the zero-axis, which suggests minimal bias and indicates that the architecture successfully captured complex nonlinear hydrochemical interactions without overfitting.vIn addition to these mathematical findings, the model's physical interpretation consistently identified Chromium (Cr^{2+}) and Fluoride (F^{-}) as the dominant contributors to the Water Quality Index (WQI), reinforcing the significance of heavy metals and ionic concentrations in the hydrochemical settings of Tamil Nadu.

Table 2: Model Evaluation Metrics

Model	MAE	RMSE	R²	Cross-validated MAE	Cross-validated RMSE	Cross-validated R²
Random Forest	0.211	0.274	0.918	0.327 ± 0.028	0.564 ± 0.041	0.742 ± 0.036
XGBoost	0.199	0.352	0.864	0.339 ± 0.031	0.589 ± 0.047	0.716 ± 0.042
LightGBM	0.267	0.358	0.860	0.346 ± 0.034	0.601 ± 0.053	0.704 ± 0.049
CatBoost	0.213	0.264	0.924	0.318 ± 0.025	0.548 ± 0.038	0.758 ± 0.031
Stacked Model	0.184	0.238	0.938	0.301 ± 0.021	0.521 ± 0.033	0.781 ± 0.027




Random Forest Feature Importance
XGBoost Feature Importance
LightGBM Feature Importance
CatBoost Feature Importance
Combined Feature Importance from All Base Models
Features
Feature Importance
Ca+Mg_vs_Na+K
SO4/Cl
Ca/Mg
Na/Cl
EC/TDS
Outlier
Fluoride (mg/L)
Cr (mg/L)
Fe (mg/L)
Ammonia (mg/L)
HCO3 (mg/L)
Cl (mg/L)
SO4 (mg/L)
K (mg/L)
Na (mg/L)
Mg (mg/L)
Ca (mg/L)
TDS (mg/L)
Turbidity (NTU)
EC (µS/cm)
pH
Random Forest
XGBoost
LightGBM
CatBoost

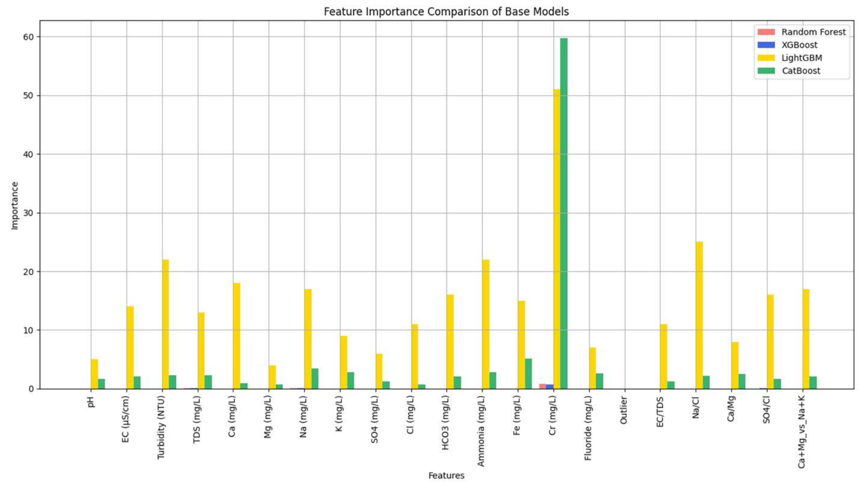


Fig.5. Feature importance for Random Forest, XGBoost, LightGBM, and CatBoost models.

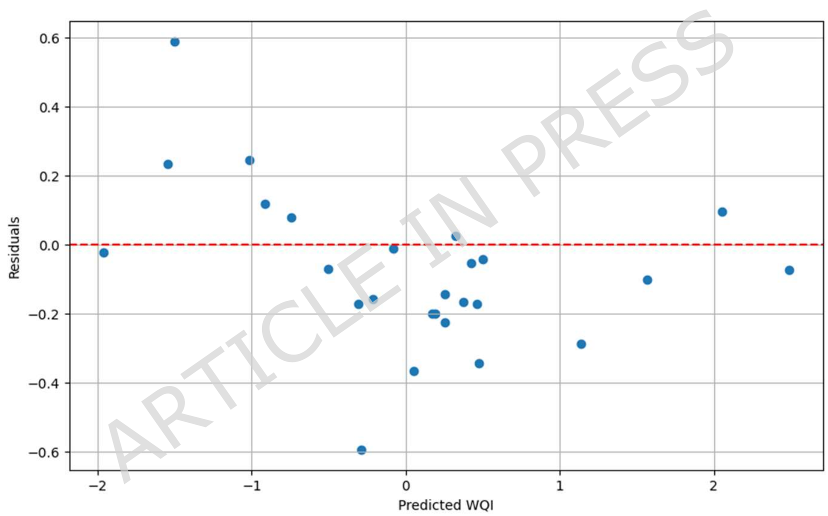


Fig.6. Residual plot for the stacked model showing predicted vs residual values.

3.4. Model Optimization

Hyperparameter optimization was conducted to enhance predictive accuracy and improve model generalization. Optuna was employed as the optimization framework, with RMSE defined as the objective function. To prevent overfitting and ensure strength, 5-fold cross-validation, early stopping, and regularization (L1/L2 penalties where applicable) were incorporated. Key hyperparameters tuned across models included max_depth, n_estimators, learning_rate, subsample ratio, and regularization coefficients. The optimization process aimed to achieve an optimal balance between model complexity and bias–variance trade-off. As presented in Table 3, substantial improvements were observed after optimization. Random Forest exhibited approximately 40% reduction in MAE and 38% reduction in RMSE, accompanied by nearly 10% improvement in R^2, indicating enhanced predictive consistency. XGBoost demonstrated around 43% reduction in MAE and 30% reduction in RMSE, with an increase of approximately 7% in R^2. LightGBM showed moderate gains, with roughly 24–25% reduction in error metrics and nearly 5% improvement in R^2. CatBoost achieved one of the most notable enhancements among base learners, with approximately 39% reduction in MAE, 45% reduction in RMSE, and about 13% increase in R^2, reflecting effective learning after parameter refinement. The learning curve of the optimized Random Forest model (Fig. 7) indicates that training and validation scores converge with increasing sample size, maintaining a narrow gap and confirming controlled variance without evidence of overfitting. The hyperparameter importance analysis (Fig. 8) reveals that max_depth had the strongest influence on performance improvement, followed by n_estimators and minimum sample parameters, emphasizing the importance of controlling tree complexity. The optimization history (Fig. 9) demonstrates a rapid early decline in RMSE, followed by gradual stabilization, confirming the efficiency of Optuna in navigating the hyperparameter space. The parallel coordinate plot (Fig. 10) further illustrates that lower objective values are associated with specific combinations of depth and estimator parameters. Following individual model optimization, a stacked ensemble model was developed. The stacked model achieved the greatest overall enhancement, with nearly 47–48% reduction in MAE and RMSE and approximately 12% improvement in R^2 compared to its baseline performance. In addition, cross-validation results indicate 5–13% superior generalization performance relative to individual models. These findings confirm that systematic hyperparameter optimization combined with ensemble stacking significantly improves predictive reliability for Water Quality Index estimation.

Table 3. Evaluation metrics comparison Before and After Optimization

Model	MAE (Before)	RMSE (Before)	R^2 (Before)	MAE (After)	RMSE (After)	R^2 (After)	Cross-validated MAE	Cross-validated RMSE	Cross-validated R^2
Random Forest	0.350	0.442	0.832	0.211	0.274	0.918	0.327 ± 0.028	0.564 ± 0.041	0.742 ± 0.036
XGBoost	0.350	0.500	0.810	0.199	0.352	0.864	0.339 ± 0.031	0.589 ± 0.047	0.716 ± 0.042
LightGBM	0.350	0.480	0.820	0.267	0.358	0.860	0.346 ± 0.034	0.601 ± 0.053	0.704 ± 0.049
CatBoost	0.350	0.480	0.820	0.213	0.264	0.924	0.318 ± 0.025	0.548 ± 0.038	0.758 ± 0.031
Stacked Model	0.350	0.460	0.835	0.184	0.238	0.938	0.301 ± 0.021	0.521 ± 0.033	0.781 ± 0.027

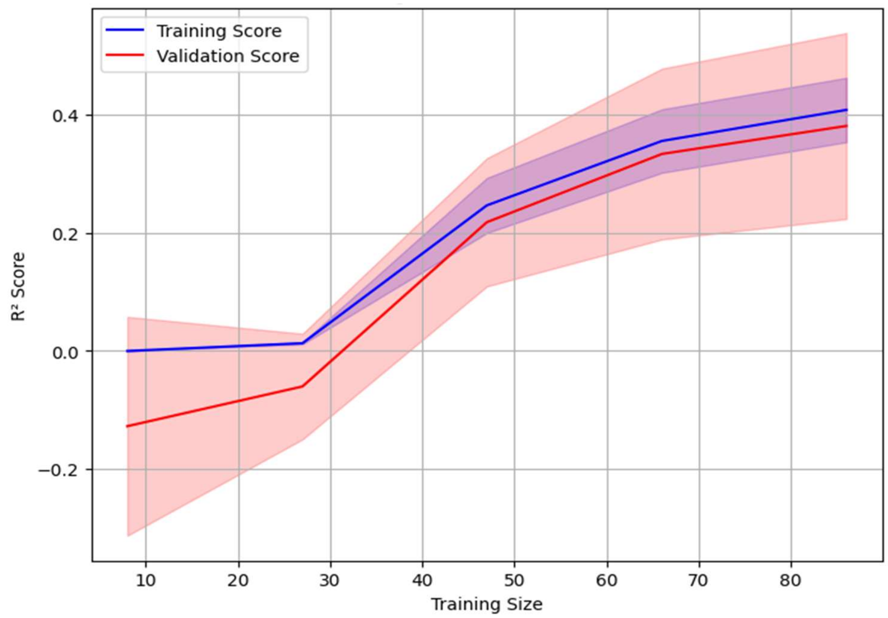


Fig.7. Learning curve

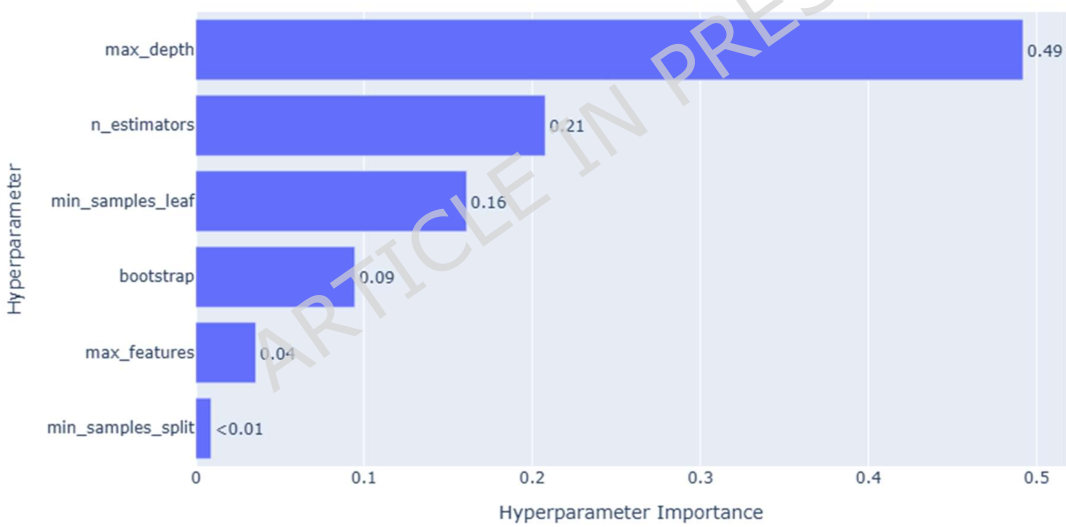


Fig.8. Hyperparameter Importance Plot

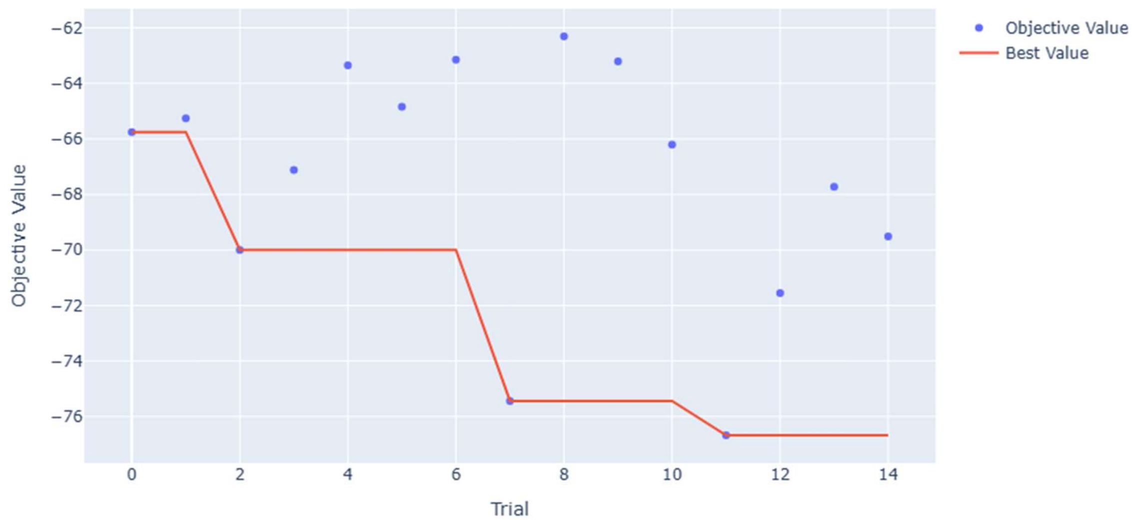


Fig.9. Optimization History Plot

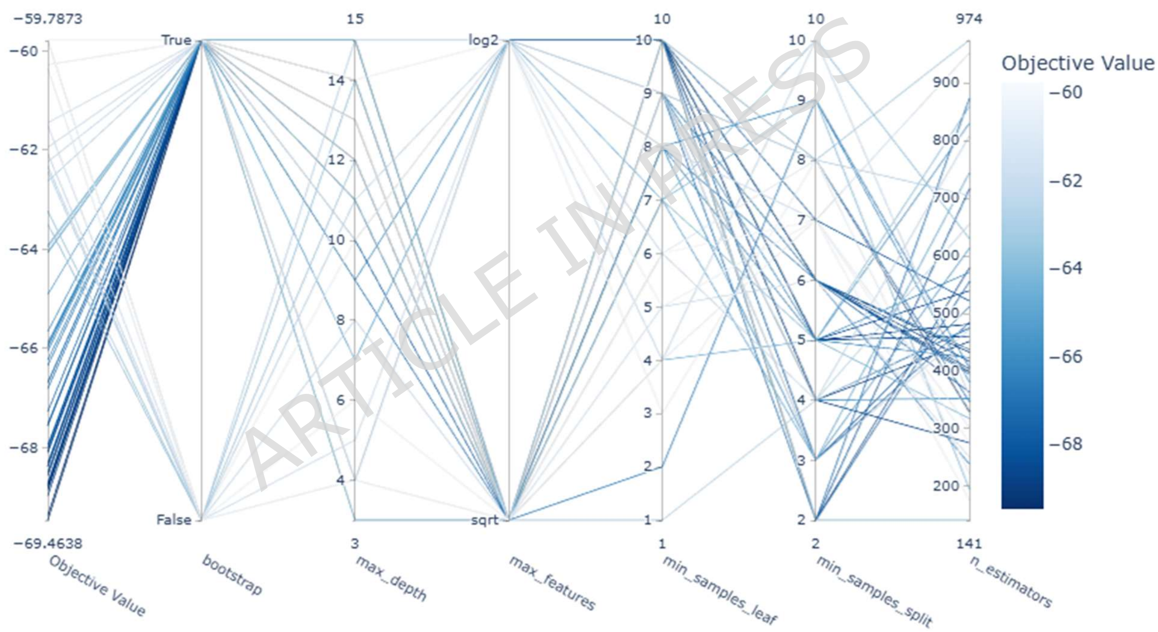


Fig. 10. Parallel Coordinate Plot

3.5. Step 5: Explainability and Sensitivity Analysis

This step presents the explainability and sensitivity analysis of the optimized stacking model to evaluate feature-level contributions to Water Quality Index (WQI) prediction. Global and local explainability techniques, including SHAP, PFI, PDP, and LIME, were used to interpret model behavior and improve transparency. Permutation Feature Importance (PFI) (Fig.11) revealed that chromium (Cr), ammonia (NH_3), fluoride, and iron (Fe) are the most influential parameters affecting WQI. This dominance indicates strong anthropogenic influence, particularly from industrial discharge (Cr, Fe) and agricultural or domestic sources (NH_3), highlighting their critical role in groundwater contamination**.** In contrast, TDS, turbidity, and pH showed relatively lower importance but contribute indirectly to ionic balance and buffering capacity. The Partial Dependence Plot (PDP) (Fig.12) demonstrated nonlinear relationships, where WQI increases sharply at lower pollutant concentrations and stabilizes at higher levels, indicating threshold-dependent hydrochemical behavior and potential equilibrium conditions within the aquifer system. The SHAP summary plot (Fig.13) provided global interpretation, showing strong positive contributions from Cr, Fe, and NH_3, confirming their role in water quality degradation. These results reflect pollutant-driven deterioration processes, while negative contributions from HCO_3^- and pH indicate natural buffering mechanisms that stabilize groundwater chemistry and reduce metal mobility. The SHAP waterfall and force plots (Figs.14–16) further explained individual predictions by illustrating feature interactions. Cr and turbidity were dominant factors increasing WQI, whereas ammonia and chloride contributed to quality deterioration. This interaction reflects the combined effect of heavy metal contamination and dissolved ionic constituents, capturing realistic hydrochemical dynamics rather than isolated statistical effects. The integration of SHAP, PFI, PDP, and LIME provides scientifically meaningful insights by linking model predictions with pollution sources, geochemical interactions, and natural attenuation processes. This enhanced interpretability supports effective groundwater monitoring and informed environmental management decisions.

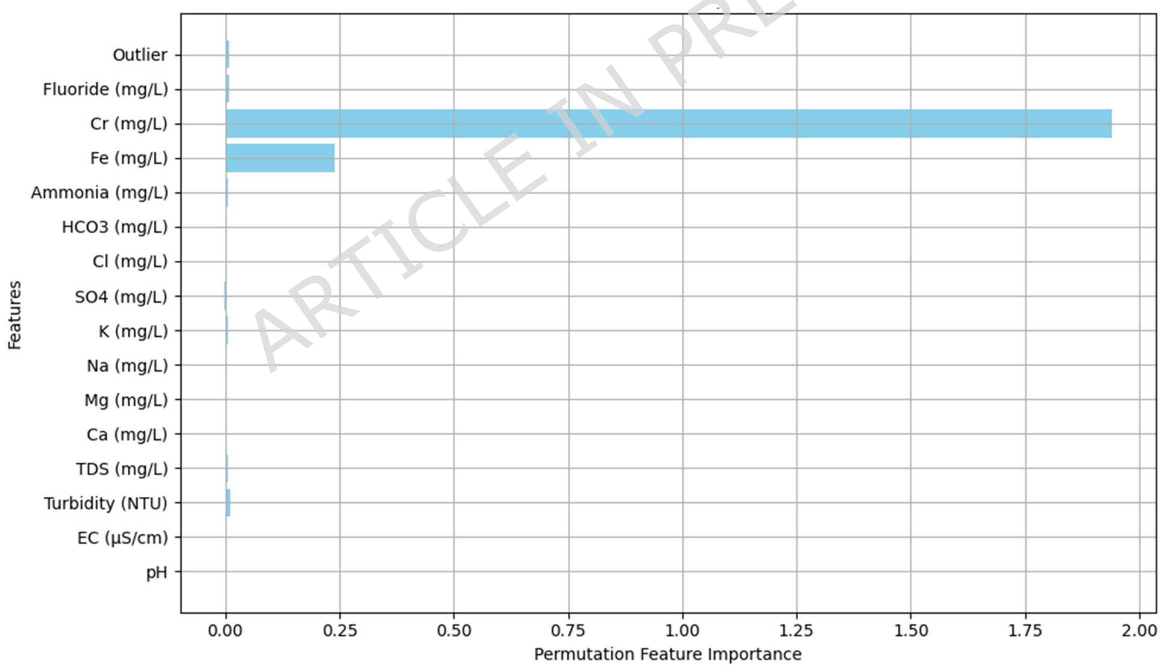


Fig.11. Permutation Feature Importance (PFI) showing ranked sensitivity of features affecting WQI prediction.

Fig.12. Partial Dependence Plot (PDP) illustrating marginal influence of a single parameter on predicted WQI.

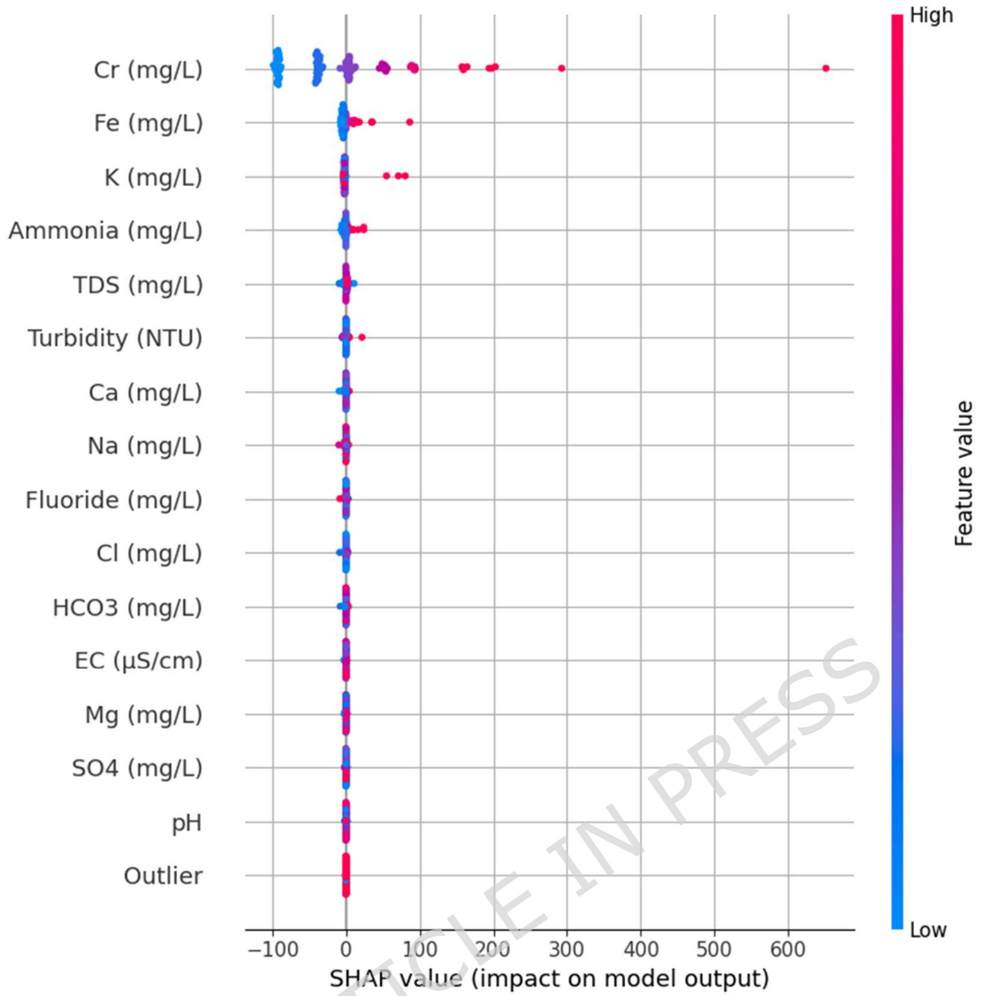


Fig.13. SHAP summary plot visualizing global contribution and directional effect of features on model predictions.

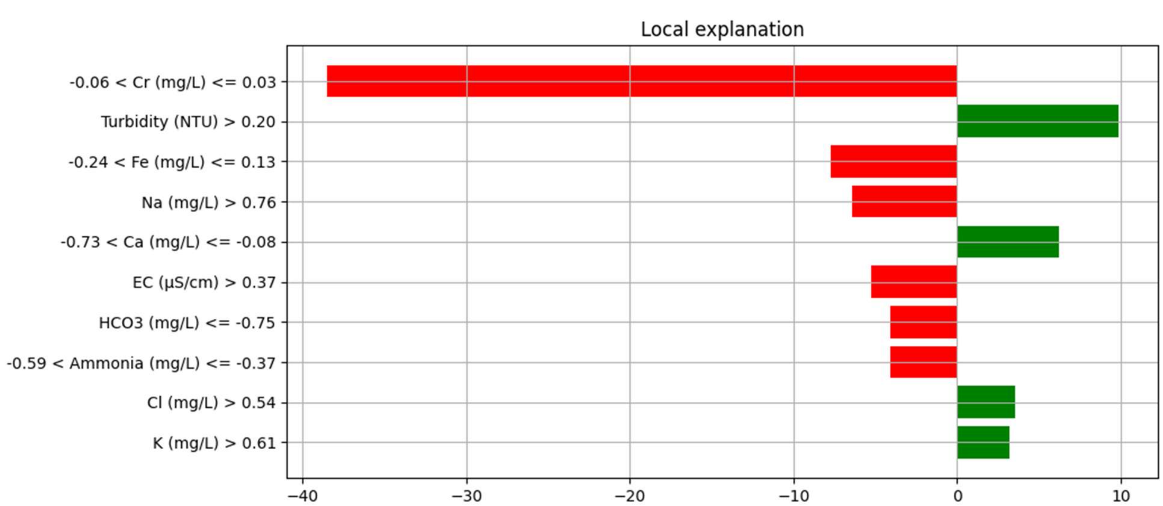


Fig.14. LIME local explanation plot highlighting instance-level feature contributions toward the predicted WQI.

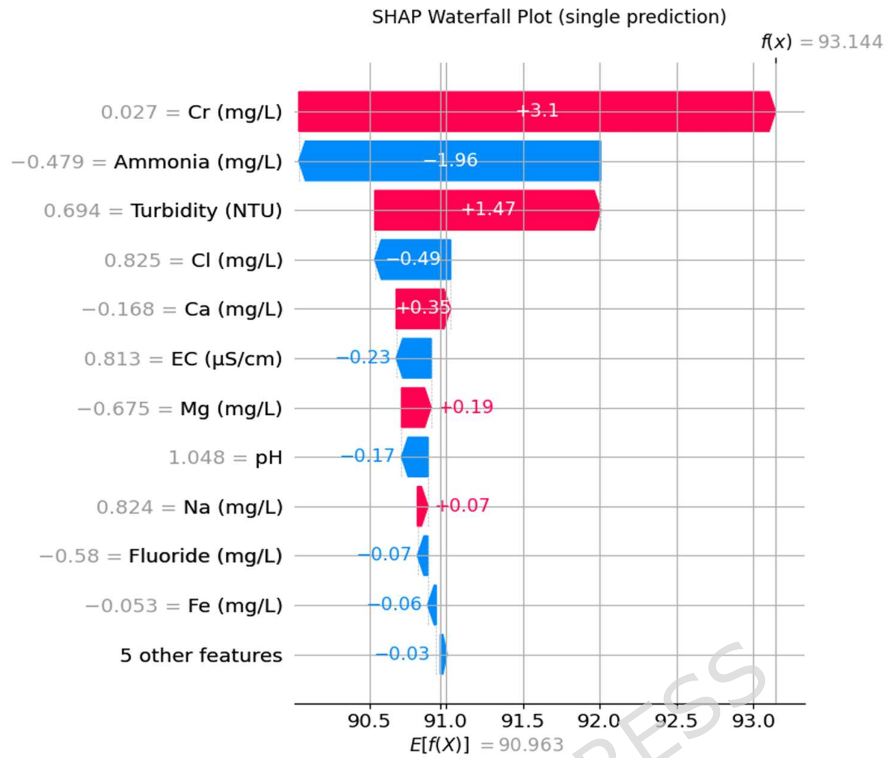


Fig.15. SHAP waterfall plot depicting the cumulative influence of individual features for a single prediction.

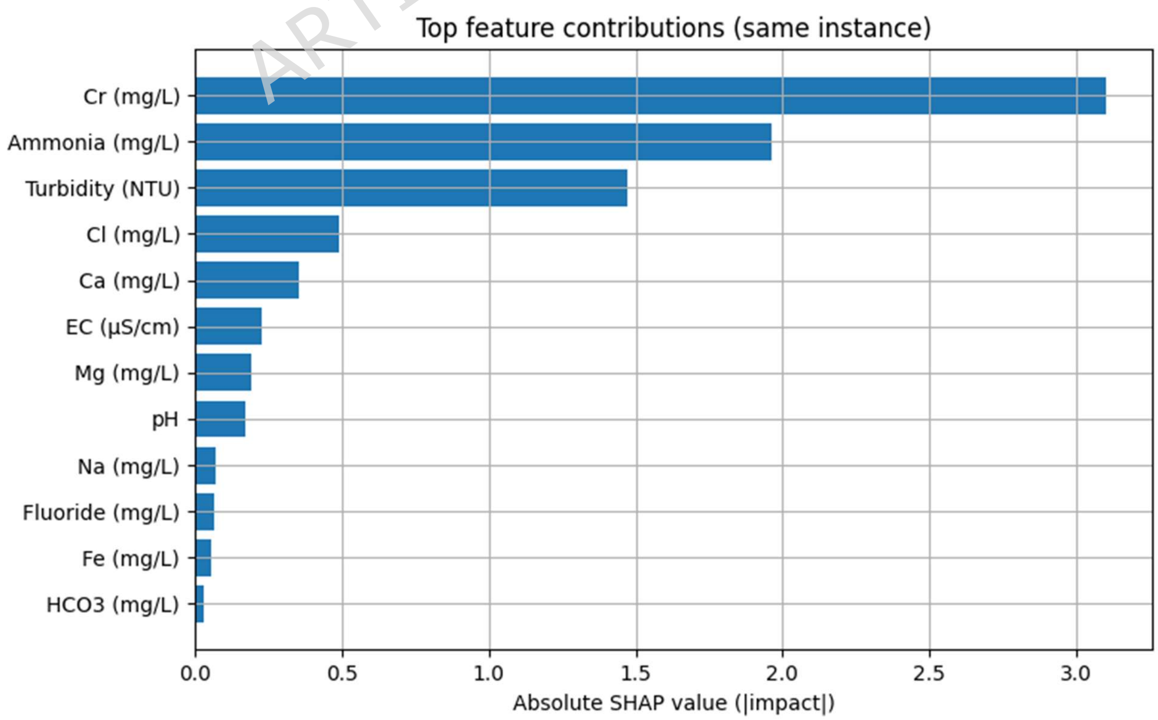


Fig.16. SHAP top feature contribution plot highlighting the most impactful features in a specific instance.

Fig.17. SHAP force plot visualizing feature interactions and opposing forces influencing the final WQI prediction.

3.6. Model Validation and Interpretability of the Optimized Stacked Ensemble

The optimized stacked ensemble model was validated to ensure predictive strength, generalization, and hydrological consistency. Performance metrics (MAE, RMSE, R^2) and cross-validation results (Table 3) confirm superior accuracy, with the stacked model achieving $R^2 = 0.938$, RMSE = 0.238, and MAE = 0.184, representing ~47–48% error reduction and ~12% improvement in R^2 over baseline models. Cross-validation statistics ($R^2 = 0.781 \pm 0.027$) indicate stable generalization with low variance, demonstrating controlled model complexity despite the limited dataset. The ±10% and ±20% error band scatter plot (Fig. 18) shows strong agreement between observed and predicted values, with most points clustered around the 1:1 line, indicating minimal bias across pollution levels. Cross-site validation further demonstrated spatial effectiveness, with only a moderate reduction in performance when applied to unseen locations, confirming the model's ability to generalize across heterogeneous hydrochemical conditions. This performance gap reflects natural spatial variability rather than model instability, supporting the framework's transferability. Global interpretability using PFI (Fig. 19) and SHAP plots (Figs. 20–21) identified chromium (Cr), TDS, ammonia, and iron as dominant predictors, consistent with hydrochemical processes such as heavy metal contamination and ionic strength variation. Negative contributions from pH and HCO_3^- indicate buffering effects. PDP and ICE plots (Fig. 22) reveal nonlinear, threshold-dependent behavior for parameters such as TDS and Ca, reflecting hydrochemical equilibrium conditions. Local explanations using SHAP force plots (Fig. 23) and LIME (Fig. 24) confirm that predictions are governed by meaningful ionic interactions rather than spurious correlations. Overall, the model demonstrates strong accuracy, spatial generalization, and interpretable insights, supporting its applicability for groundwater quality assessment and decision-support systems.

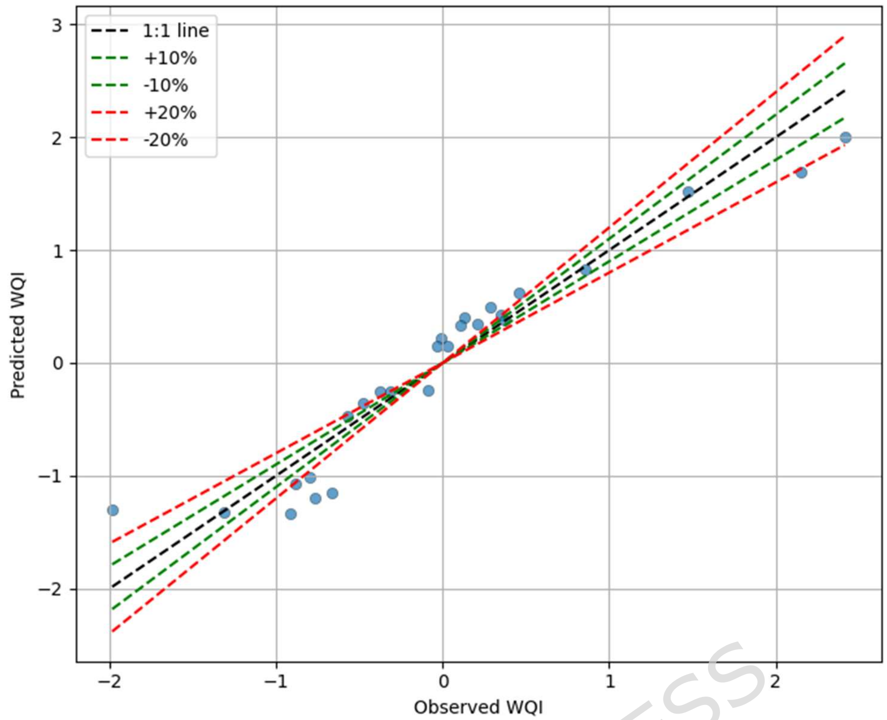


Fig. 18. ±10 % and ±20 % scatter plot for model predictions showing alignment of observed and predicted WQI values along the 1:1 ideal line.

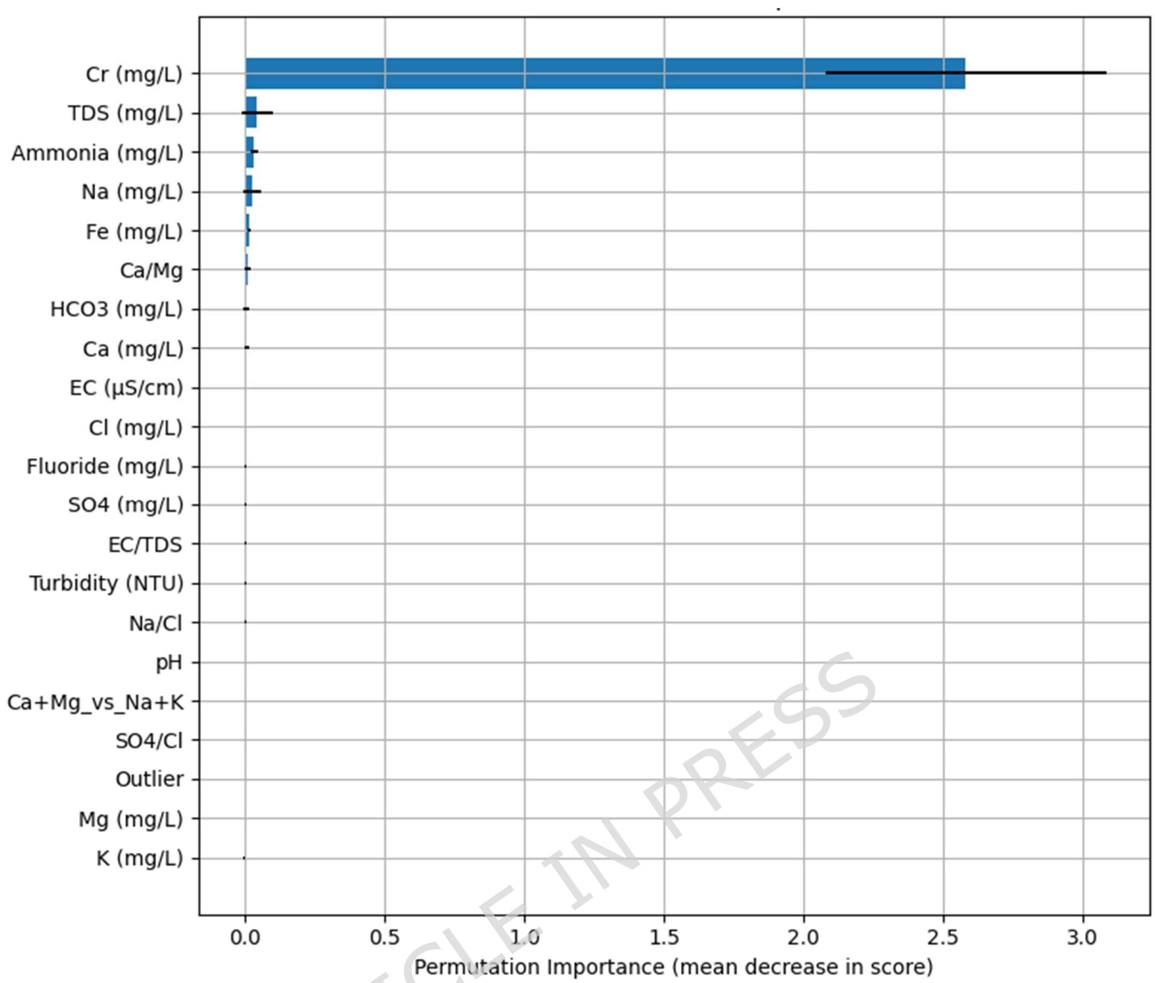


Fig. 19. Global Permutation Feature Importance (PFI) plot showing the dominant influence of Cr and TDS on model performance.

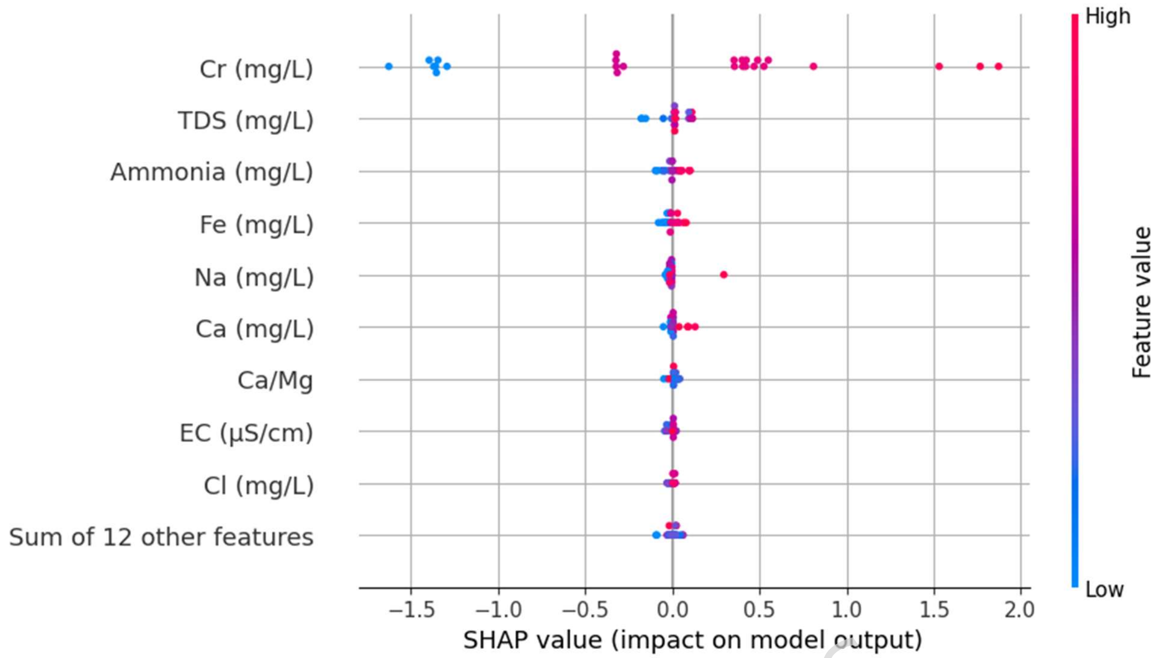


Fig. 20. SHAP summary (beeswarm) plot depicting global feature impact and direction of influence on WQI.

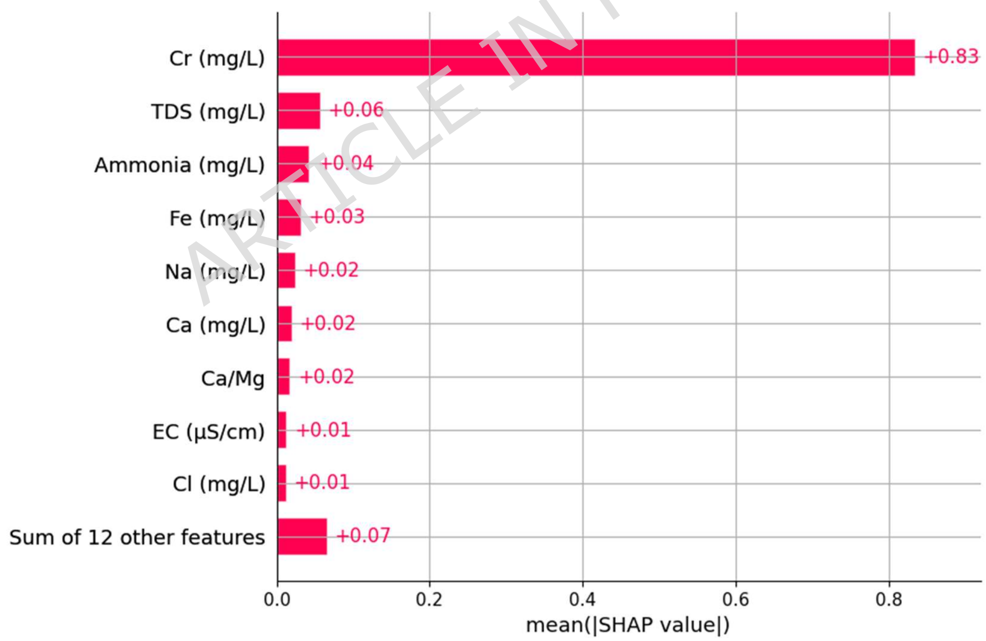


Fig. 21. Mean |SHAP| bar plot illustrating average magnitude of feature importance across samples.

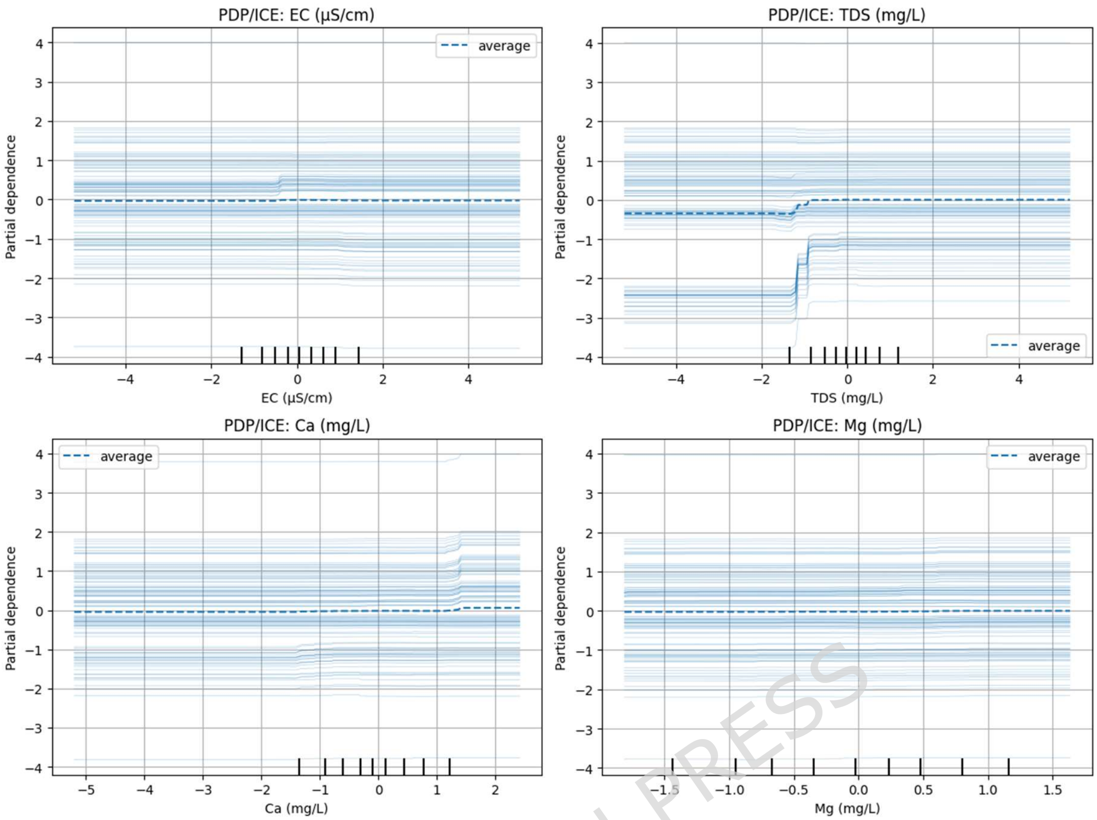


Fig. 22. Partial Dependence and Individual Conditional Expectation (PDP/ICE) plots for EC, TDS, Ca, and Mg highlighting nonlinear threshold responses.

Fig. 23. SHAP force plot for a single sample explaining individual feature contributions to prediction deviation from the baseline.

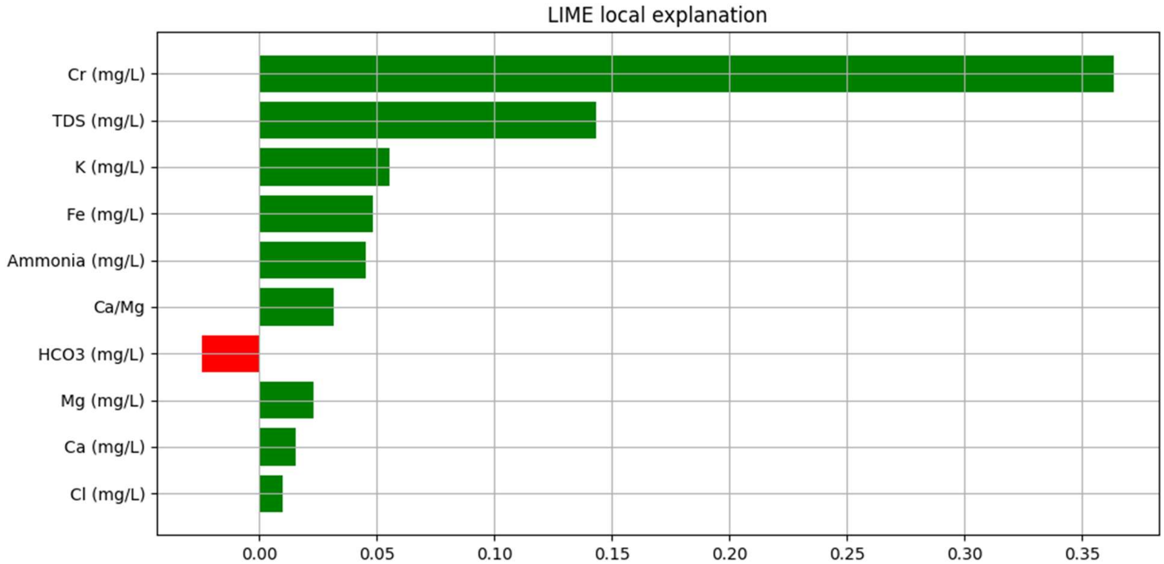


Fig. 24. LIME local explanation plot showing sample-specific positive and negative feature impacts on the predicted WQI.

3.7 Step 7: Visualization, Interpretation and Policy Recommendations

The predictive intelligence of the model was converted into understandable visual and policy-level insights during the last analytical stage (Table 4). To investigate the geographical patterns, variable influence, and pollutant dominance influencing Water Quality Index (WQI) variance among sites, a series of visualizations (Figs. 25–27) and explainable AI metrics were utilized. Dasanike Kere, Chikka Kere, and Karihobanahalli Lake have exceptionally high WQI values (> 700), which shows a clear geographical difference. However, the values at upstream and semi-rural sites are significantly lower, highlighting the fact that anthropogenic intensity is what drives spatial variation. Instead of universal regulation, this differential mapping emphasizes the need for zone-specific mitigation. The correlation heatmap (Fig. 25) shows WQI parametric linkages. The most common contaminants were heavy metals, including chromium (Cr) (r = 0.968), iron (Fe) (r = 0.291), and ammonia (NH_3) (r = 0.076). In less-stressed zones, weak or negative correlations validate the buffering roles of HCO_3^-, Mg^{2+}, and Ca^{2+}, indicating ionic equilibrium. As heavy-metal enrichment and fertilizer input replace natural ionic balance in urbanized catchments, The mean |SHAP| important plot (Fig. 26) and SHAP summary (Fig. 27) rank parameters by predictive power. Cr (mg/L) leads with a mean |SHAP| of 51.79, followed by Fe (4.15), Ammonia (2.01), K (1.56), and Turbidity (0.79). These traits imply quality decline and raise WQI. The stabilizing effect of HCO_3^-, Ca^{2+}, and Mg^{2+} is shown by their slightly negative SHAP contributions. All these data show that the ensemble model found chemically logical patterns that make hydrological sense.

Table 4. Feature Importance (SHAP) and Correlation with WQI

Feature	Mean|SHAP|	Corr(WQI)
Cr (mg/L)	51.788	0.968
Fe (mg/L)	4.1453	0.291
Ammonia (mg/L)	2.015	0.075

K (mg/L)	1.557	0.128
Turbidity (NTU)	0.793	0.046
TDS (mg/L)	0.615	-0.227
HCO3 (mg/L)	0.467	-0.028
Na (mg/L)	0.389	0.069
SO4 (mg/L)	0.294	-0.111
Mg (mg/L)	0.293	-0.089
Ca (mg/L)	0.290	-0.034
Fluoride (mg/L)	0.246	0.238
EC (ÂµS/cm)	0.230	-0.071
Cl (mg/L)	0.205	-0.042
pH	0.161	0.111
Outlier	0.092	-0.321

Cr, Fe, Ammonia, K, and Turbidity are the main variables affecting WQI elevation, according to correlation and SHAP diagnostics. A positive connection and high SHAP magnitude imply direct deterioration from suspended particles (turbidity), home effluents (ammonia), and industrial discharges. For alkaline earth metals and HCO_3^-, negative correlations suggest self-regulation via dilution and buffering. The ensemble model characterizes natural attenuation processes and pollutant load with hydrologic realism.

3.7.1. Policy Implications

The findings provide a numerical basis for environmental governance:

1. Minimize the Cr emissions through observing zero-discharged regulations and carrying out the necessary audits on the industrial effluent.

2. Aeration and filtration should be used retrofit in controlling iron and corrosion of water distribution systems.

3. Manage nutrients and biotreatment to reduce the quantities of ammonia released at home and agricultural sources.

4. Manage salinity and TDS in high-EC regions by using methods such as blending, recharging, or partial desalination.

5. Suspended particles and turbidity are treated by means of improved coagulation-flocculation units.

6. Keep buffering ions (HCO3-, Ca2+, and Mg2+) to level the PH and limit the solubility of the metals.

7. In order to rank high-risk sites with composite risk scores (mean |SHAP| x exceedance frequency) to prioritize inspection schedules and remedial efforts. This phase creates a clear, policy-ready framework by combining the analytical process. This phase connects data science and environmental management by combining statistical correlation, interpretability based on XAI, and geographical visualization. The strategy turns predictive

knowledge into sustainable decision support by empowering regulators to identify, pinpoint, and address the most significant factors causing deterioration.

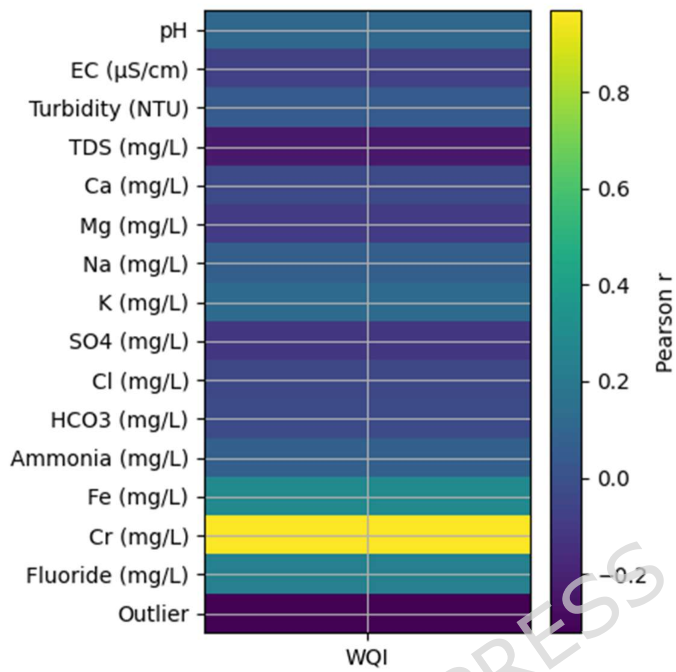


Fig. 25. Correlation heatmap between WQI and physicochemical parameters; Cr, Fe, and Ammonia exhibit strong positive associations.

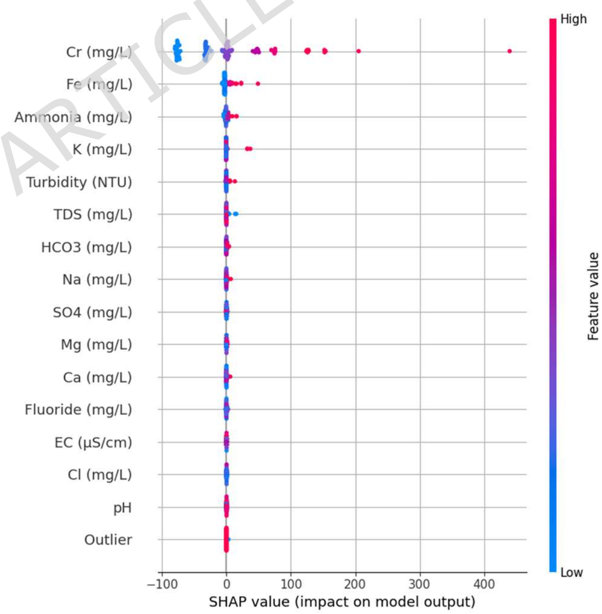


Fig. 26. SHAP summary (beeswarm) plot showing global feature contributions and directionality of influence on WQI.

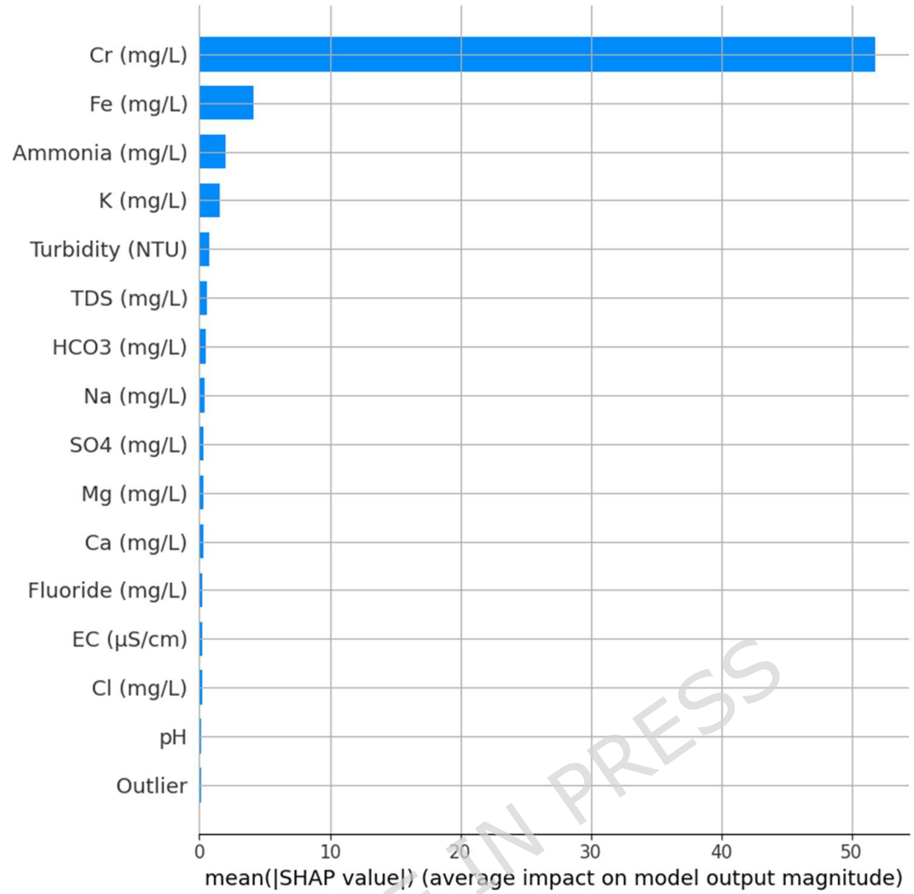


Fig. 27. Mean |SHAP| bar plot ranking predictors by average impact magnitude, confirming Cr as the dominant driver.

4. Conclusion

This study developed an interpretable artificial intelligence framework for groundwater quality prediction and decision support by integrating hybrid ensemble learning, data analytics, and explainable artificial intelligence techniques. Using groundwater data collected between 2015 and 2025, physicochemical parameters from multiple monitoring locations were pre-processed through dimensionality reduction, normalization, and outlier detection to ensure reliable model training.

- The hybrid ensemble architecture integrating Random Forest, XGBoost, and CatBoost with a meta-regressor demonstrated strong predictive performance, achieving $R^2 = 0.92$, RMSE = 0.27, MAE = 0.21, and Nash–Sutcliffe Efficiency (NSE) = 0.92, indicating high predictive reliability and hydrological consistency.
- Model validation confirmed robustness across diverse hydrogeochemical environments, with most predicted Water Quality Index (WQI) values falling within ±10.00% of observed measurements, demonstrating strong generalization capability.

- Explainable Artificial Intelligence analyses using SHAP, LIME, Permutation Feature Importance, and PDP/ICE identified the most influential hydrochemical parameters controlling groundwater quality.
- Heavy metals such as iron (Fe) and chromium (Cr), along with ammonia, total dissolved solids (TDS), and turbidity, were identified as major contributors to WQI degradation in industrial and urban catchments.
- Hydrochemical buffering ions including HCO_3^-, Ca^{2+}, and Mg^{2+} were found to contribute positively to groundwater stabilization, highlighting their role in natural attenuation processes.
- The framework successfully translates complex machine learning outputs into interpretable environmental insights, supporting evidence-based decision making for pollution control and groundwater management.
- Policy-relevant recommendations include regulation of industrial metal discharge, monitoring of ammonia and salinity levels, preservation of buffering ions, and adoption of AI-assisted monitoring systems for sustainable groundwater governance.

4.1 Future Research Directions and limitations

Although the present study demonstrates the effectiveness of an interpretable AI framework for groundwater quality prediction and decision support, the current work represents only an initial step toward fully intelligent and adaptive water quality management systems. Further research is required to enhance the methodological depth, scalability, and real-time applicability of the proposed framework. In particular, future studies should investigate the integration of advanced spatiotemporal deep learning architectures, such as ConvLSTM and Graph Neural Networks (GNNs), to better capture complex seasonal variability and spatial dependencies in hydrochemical parameters. The incorporation of Internet of Things (IoT)-based sensor networks and edge computing platforms is also necessary to enable continuous data acquisition and near real-time calibration of Water Quality Index (WQI) prediction models. Additionally, coupling the current machine learning framework with hydrodynamic or watershed simulation models would improve the understanding and prediction of pollutant transport mechanisms under varying climatic and anthropogenic pressures. Future work should also focus on incorporating uncertainty quantification, probabilistic modelling, and multi-objective optimization to strengthen risk-based environmental decision-making and resource allocation strategies. Finally, translating predictive outputs into operational decision-support systems through cloud-based visualization platforms and AI-driven policy dashboards will be essential for bridging the gap between research development and practical implementation in environmental governance. These advancements will be critical for transforming the current framework into a comprehensive, scalable, and adaptive tool for sustainable groundwater quality monitoring and management.

A limitation of the present study is the use of the Water Quality Index (WQI) as the target variable. Since WQI is a composite index derived from multiple physicochemical parameters using predefined weights and aggregation rules, the model inherently inherits uncertainties associated with its formulation. Variations in weighting schemes or classification thresholds may influence prediction outcomes. While WQI provides a simplified and policy-relevant indicator for groundwater quality assessment, it may mask the behavior of individual parameters. Future research may focus on predicting individual hydrochemical parameters alongside WQI to achieve more detailed process-level understanding.

Declaration of Competing Interest:

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement – The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Funding: This project have not received funding from any of the agencies

Author Contribution:

G.Shyamala: Problem findings, experimental work, writing manuscript.

Prakhash Neelemagam : Supervising, supporting for writing article and review of manuscript.

Belin Jude Alphonse and Missgna Addisalem Berhe :Review and Final drafting

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